

Mathematical background
Branching, quasi-stationarity and absorption models

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1 Introduction

Every model in this thesis describes a population small enough that chance belongs to the mechanism rather than to the error bars. A few individuals founding a colony, or a few particles held inside a single compartment that may later open: in each case the count is far from the regime in which a deterministic rate equation carries useful information, and the questions worth asking — will the population establish itself at all, and if it survives, how large will it be? — are questions about a distribution rather than about a mean. This chapter assembles the apparatus needed to answer them. The treatment is deliberately abstract, in counts, rates and generating functions, and the modelling motifs that motivate it are left to the chapters that use them.

Two threads run through the material, and it is worth naming both before either is developed.

The first is *conditioning on survival*. A subcritical branching process dies out with probability one, so its unconditional mean decays geometrically and reports almost nothing about the realisations one would actually observe. Restrict attention to those realisations still alive at generation n , however, and a different picture appears: the conditional population settles onto a fixed distribution — the *quasi-stationary* distribution — whose mean is a constant. For the binary Galton–Watson process with replication probability $p < \frac{1}{2}$ that constant is $1/A(p)$, where

$$A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n}$$

and S_n is the probability of survival to generation n . A large part of this chapter is an account of $A(p)$: what it is, how to compute it, and why, despite a considerable search, no elementary formula for it has been found.

The second thread is *the near-critical regime*. As p rises towards $\frac{1}{2}$ the process takes ever longer to die, the quasi-stationary mean $1/A(p)$ diverges, and any numerical scheme that iterates to stationarity becomes impractical. Just above $\frac{1}{2}$ the size of the founding cohort begins to weigh heavily on whether the process survives at all. The behaviour on either side of $p = \frac{1}{2}$ is what makes small founding numbers interesting, and it is also what makes them awkward to compute with.

[Section 2](#) sets out the standard machinery the rest of the chapter uses: continuous-time Markov chains and their generators, exponential clocks and the competition between them, generating functions, and the linear birth–death process solved in closed form. Readers already fluent in that material may read it as a fixing of notation. [Section 3](#) introduces the Galton–Watson process and establishes the two facts on which everything later rests — that extinction is certain at and below criticality, and that at criticality the expected lifetime is nonetheless infinite. [Section 4](#) conditions on survival, derives the limiting conditional mean and variance, and so introduces $A(p)$. [Section 5](#) asks what changes when the process starts from a cohort of k individuals rather than one, records a discrete birth–death–catastrophe calculation of the kind developed properly in later chapters, and works out the corresponding constant for the continuous-time birth–death process, where a closed form *is* available. [Section 6](#) is devoted to $A(p)$ itself: an infinite-product representation, an exact series with two-sided bounds that pin the constant between the continuous-time answer and twice it, the near-critical asymptotic $A(p) \sim 2(1 - 2p)$, and an account of why the exact function resists closed-form description — a question that turns out to concern the Koenigs linearisation of the logistic map. [Section 7](#) then changes tack and develops generating-function partial differential equations and the method of characteristics for the absorption models used later in the thesis.

Standard results are presented as standard. Where an argument is heuristic, or where the numerics are doing the real work, the text says so at that point rather than in a general disclaimer, and every claim that can be checked numerically has been.

2 The tools

The models of this thesis are built from three ingredients: a count that jumps by one, a clock that decides when it jumps, and a generating function that turns an infinite system of differential equations into a single equation one can hope to solve. This section fixes each in turn, in the order in which one needs them, and closes with the linear birth–death process — the continuous-time reference model against which every later construction is measured.

2.1 Exponential clocks and their competition

Waiting times enter first, because everything else is assembled from them. A non-negative random variable T is exponentially distributed with rate $\lambda > 0$, written $T \sim \text{Exp}(\lambda)$, when it has density and distribution function

$$f_T(x) = \lambda e^{-\lambda x}, \quad F_T(x) = \Pr\{T \leq x\} = 1 - e^{-\lambda x} \quad (x \geq 0), \quad (1)$$

with $\mathbb{E}(T) = 1/\lambda$ and $\text{Var}(T) = 1/\lambda^2$. The rate is the reciprocal of the mean waiting time, and it is the parameter in which models are specified.

The property that earns the exponential law its place is that waiting confers no advantage.

Definition 2.1 (Memorylessness). A non-negative random variable T is memoryless if

$$\Pr\{T > t_1 + t_2 \mid T > t_1\} = \Pr\{T > t_2\} \quad \text{for all } t_1, t_2 > 0. \quad (2)$$

For $T \sim \text{Exp}(\lambda)$ this is immediate from (1), since $\Pr\{T > x\} = e^{-\lambda x}$ and therefore

$$\Pr\{T > t_1 + t_2 \mid T > t_1\} = \frac{e^{-\lambda(t_1+t_2)}}{e^{-\lambda t_1}} = e^{-\lambda t_2} = \Pr\{T > t_2\}.$$

Among continuous laws the exponential is the only one with this property, which is why a model whose event rates do not depend on elapsed time is an exponential-clock model whether or not it was described as one.

Populations offer several events at once — a birth, a death, an absorption, a catastrophe — so what matters in practice is not a single clock but a race between them. Take independent clocks $T_1 \sim \text{Exp}(\lambda_1)$ and $T_2 \sim \text{Exp}(\lambda_2)$. Conditioning on the value of T_2 and integrating,

$$\Pr\{T_1 < T_2\} = \int_0^\infty \left(\int_0^y \lambda_1 e^{-\lambda_1 x} dx \right) \lambda_2 e^{-\lambda_2 y} dy = \int_0^\infty (1 - e^{-\lambda_1 y}) \lambda_2 e^{-\lambda_2 y} dy = \frac{\lambda_1}{\lambda_1 + \lambda_2}, \quad (3)$$

while for the time of the first event

$$\Pr\{\min\{T_1, T_2\} > x\} = \Pr\{T_1 > x\} \Pr\{T_2 > x\} = e^{-(\lambda_1 + \lambda_2)x}, \quad (4)$$

so that $\min\{T_1, T_2\} \sim \text{Exp}(\lambda_1 + \lambda_2)$. The same two calculations run unchanged for n independent clocks with rates $\lambda_1, \dots, \lambda_n$. Writing $\Lambda = \sum_i \lambda_i$ for the total rate,

$$\min_i T_i \sim \text{Exp}(\Lambda), \quad \Pr\left\{T_i = \min_j T_j\right\} = \frac{\lambda_i}{\Lambda}, \quad (5)$$

and the two are independent: knowing which channel fired first tells one nothing about when it fired. Rates therefore do two separate jobs, setting the pace of events through Λ and apportioning them among channels through the ratios λ_i/Λ .

Remark 2.2 (Simulation). (5) is also an algorithm. From the current state, compute the rate a_k of each available event and their total Λ ; draw a holding time $\tau \sim \text{Exp}(\Lambda)$; select event k with probability a_k/Λ ; update the state and advance the clock by τ . This is Gillespie’s stochastic simulation algorithm, and it is exact for the chains defined below. Simulation is used in this chapter only to check analysis, and the figures say where.

2.2 Continuous-time Markov chains

A stochastic process is a family $\{\mathbf{X}_t : t \geq 0\}$ of random variables on a common probability space. Throughout this thesis the index set is $[0, \infty)$ and the state space is a subset of $\{0, 1, 2, \dots\}$: time runs continuously, and the state is a count. The process is a *continuous-time Markov chain* (CTMC) when its future depends on its past only through its present,

$$\Pr\{\mathbf{X}_{t_{n+1}} = i_{n+1} \mid \mathbf{X}_{t_0} = i_0, \dots, \mathbf{X}_{t_n} = i_n\} = \Pr\{\mathbf{X}_{t_{n+1}} = i_{n+1} \mid \mathbf{X}_{t_n} = i_n\} \quad (6)$$

for all $0 \leq t_0 < \dots < t_{n+1}$ and all states $i_0, \dots, i_{n+1}$. Memorylessness is what buys (6): if every pending event is timed by an exponential clock, the elapsed time carries no information, and the present state is a sufficient summary.

Write $p_{ij}(s, t) = \Pr\{\mathbf{X}_t = j \mid \mathbf{X}_s = i\}$ for the transition probabilities. Every chain in this thesis is *time-homogeneous*, meaning that $p_{ij}(s, t)$ depends on s and t only through the lag $t - s$, so that we may write $p_{ij}(t)$ and collect these into a matrix $P(t)$. Models are not specified by writing $P(t)$ down, however, but by writing rates.

Definition 2.3 (Infinitesimal generator). The generator of a time-homogeneous CTMC is the matrix $Q = (q_{ij})$ with

$$q_{ij} = \lim_{\Delta t \rightarrow 0^+} \frac{p_{ij}(\Delta t) - \delta_{ij}}{\Delta t} \quad (i \neq j), \quad q_{ii} = -\sum_{j \neq i} q_{ij}, \quad (7)$$

whenever these limits exist; equivalently $p_{ij}(\Delta t) = \delta_{ij} + q_{ij}\Delta t + o(\Delta t)$.

Off-diagonal entries are non-negative and each row sums to zero. The generator also recovers the pathwise picture of [section 2.1](#): while the chain occupies state i it waits an exponential time of rate $\Lambda_i = -q_{ii} = \sum_{j \neq i} q_{ij}$, and it then moves to j with probability q_{ij}/Λ_i . One clock per exit channel, and (5) settles the rest. A state i with $q_{ii} = 0$ has no exits at all and is called *absorbing*; in a population model the absorbing state is $i = 0$, extinction, and the whole of this chapter is in one way or another about how a process approaches it.

Differentiating the Chapman–Kolmogorov relation gives the forward equations,

$$\frac{dp_{ij}(t)}{dt} = \sum_k p_{ik}(t) q_{kj}, \quad \text{that is} \quad \frac{dP}{dt} = P(t) Q, \quad P(0) = I. \quad (8)$$

Conventions for which system is called forward and which backward differ between texts; (8) is the one used throughout, and it is the useful one here because it reads as a balance of probability flow into and out of state j . Written for the marginal probabilities $p_j(t) = \Pr\{\mathbf{X}_t = j\}$ it is the master equation of the model, and every derivation in [section 7](#) begins by writing it down.

2.3 Probability generating functions

The master equation is an infinite system of coupled differential equations, one per state, and solving it state by state is rarely possible. The standard remedy is to pack the whole system into one analytic object.

Definition 2.4 (Probability generating function). For a random variable $\mathbf{X}$ on $\{0, 1, 2, \dots\}$ with $\Pr\{\mathbf{X} = k\} = p_k$, the probability generating function is

$$G_{\mathbf{X}}(z) = \mathbb{E}(z^{\mathbf{X}}) = \sum_{k \geq 0} p_k z^k, \quad |z| \leq 1. \quad (9)$$

The series converges on the closed unit disc because the coefficients are probabilities, and the distribution can be read back off it by differentiation. Four consequences are used constantly:

$$G_{\mathbf{X}}(1) = 1, \quad G_{\mathbf{X}}(0) = p_0, \quad p_k = \frac{1}{k!} G_{\mathbf{X}}^{(k)}(0), \quad \mathbb{E}(\mathbf{X}) = G'_{\mathbf{X}}(1). \quad (10)$$

The second of these is the one that motivates the whole apparatus for small populations: if $\mathbf{X}$ is a population size then $G_{\mathbf{X}}(0)$ is the extinction probability. The variance follows from the second derivative, $\text{Var}(\mathbf{X}) = G''_{\mathbf{X}}(1) + G'_{\mathbf{X}}(1) - (G'_{\mathbf{X}}(1))^2$.

A fifth property does the structural work. If $\mathbf{X}$ and $\mathbf{Y}$ are independent then $\mathbb{E}(z^{\mathbf{X}+\mathbf{Y}}) = \mathbb{E}(z^{\mathbf{X}})\mathbb{E}(z^{\mathbf{Y}})$, so

$$G_{\mathbf{X}+\mathbf{Y}}(z) = G_{\mathbf{X}}(z) G_{\mathbf{Y}}(z). \quad (11)$$

Independent individuals therefore multiply their generating functions, which is why generating functions and branching arguments belong together: (11) is what turns “each of the n founders behaves independently” into an n th power.

For a process rather than a single random variable the generating function acquires a time argument,

$$G(z, t) = \mathbb{E}(z^{\mathbf{X}_t}) = \sum_{j \geq 0} p_j(t) z^j, \quad (12)$$

and the extinction probability by time t is $p_0(t) = G(0, t)$. Multiplying the master equation (8) by z^j , summing over j and recognising the resulting sums as derivatives of G replaces the infinite system by a single first-order partial differential equation for $G(z, t)$. That compression is the analytical centre of this chapter. When two counts must be tracked at once — particles inside a compartment and particles outside it, say — the same construction runs with one variable per count,

$$G(x, y, t) = \mathbb{E}(x^{\mathbf{X}_t} y^{\mathbf{Y}_t}) = \sum_{i \geq 0} \sum_{j \geq 0} p_{i,j}(t) x^i y^j, \quad (13)$$

which is the object solved in [section 7](#) and the natural starting point for the multi-type models developed later in the thesis.

2.4 The linear birth–death process

Everything above can now be exercised on the model that all later constructions extend. Let $\mathbf{X}_t$ count individuals, each of which independently gives birth at rate λ and dies at rate μ . Because the individuals act independently, the population rates are linear in the state,

$$q_{i,i+1} = i\lambda, \quad q_{i,i-1} = i\mu, \quad q_{ii} = -i(\lambda + \mu) \quad (i \geq 1), \quad (14)$$

and $i = 0$ is absorbing. The master equation (8) reads

$$\frac{dp_0}{dt} = \mu p_1, \quad \frac{dp_j}{dt} = \lambda(j-1)p_{j-1} - (\lambda + \mu)j p_j + \mu(j+1)p_{j+1} \quad (j \geq 1), \quad (15)$$

and sequential solution stalls immediately, since each equation involves its neighbours. Multiplying by z^j and summing, as in [section 2.3](#), collapses the system to

$$\frac{\partial G(z, t)}{\partial t} = (\lambda z - \mu)(z - 1) \frac{\partial G(z, t)}{\partial z}, \quad G(z, 0) = z^N, \quad (16)$$

for a process started from $\mathbf{X}_0 = N$ individuals. (16) is a first-order linear equation of exactly the type solved by the method of characteristics in [section 7](#); here we simply record its solution, which for $\lambda \neq \mu$ is

$$G(z, t) = \left(\frac{\mu(1 - z) - (\mu - \lambda z)e^{(\mu - \lambda)t}}{\lambda(1 - z) - (\mu - \lambda z)e^{(\mu - \lambda)t}} \right)^N. \quad (17)$$

Two checks fix the reading. At $t = 0$ the exponentials are unity and the bracket collapses to z , so $G = z^N$; and at $z = 1$ numerator and denominator agree, so $G = 1$ and probability is conserved. The mean follows from $\mathbb{E}(\mathbf{X}_t) = \partial_z G|_{z=1}$,

$$\mathbb{E}(\mathbf{X}_t) = N e^{(\lambda - \mu)t}, \quad (18)$$

which is the deterministic answer and, on its own, an uninformative one: it reports exponential growth or exponential decay and says nothing about the chance of neither.

That information sits at $z = 0$. Setting $z = 0$ in (17), and treating the critical case $\lambda = \mu$ by taking the limit, gives the probability of extinction by time t [1],

$$p_0(t) = G(0, t) = \begin{cases} \left(\frac{\mu - \mu e^{(\mu - \lambda)t}}{\lambda - \mu e^{(\mu - \lambda)t}} \right)^N, & \lambda \neq \mu, \\ \left(\frac{\lambda t}{1 + \lambda t} \right)^N, & \lambda = \mu, \end{cases} \quad (19)$$

and the survival probability $S^{(N)}(t) = 1 - p_0(t)$. Letting $t \rightarrow \infty$ in (19) recovers the classical trichotomy: extinction is certain when $\lambda \leq \mu$, including at $\lambda = \mu$, and occurs with probability $(\mu/\lambda)^N$ when $\lambda > \mu$. A population with a per-capita growth advantage can still be lost, and the chance of losing it falls only geometrically in the founding number N .

This process is the continuous-time backbone of the thesis; later chapters add a catastrophe channel to (14) and follow the same route from rates to master equation to generating function. The next section takes the other route, replacing the exponential clock by a generational one, and asks the same questions of the resulting discrete-time process.

3 The Galton–Watson process

Replace the exponential clock of [section 2.4](#) by a generational one and the birth–death process becomes the Galton–Watson process: a discrete-time, discrete-state Markov chain in which every member of the population independently either reproduces or dies at each generational increment. Formulated by Francis Galton and Henry Watson in the late nineteenth century, it remains the most basic model of stochastic branching [2, 3], and it is the setting in which the conditioning arguments of this chapter are cleanest.

Let each individual produce, at the end of its life and independently of every other, a number of offspring distributed as the random variable ξ , with

$$\Pr \{ \xi = k \} = p_k, \quad k = 0, 1, 2, \dots,$$

and write $\mu = \mathbb{E}(\xi)$ for the mean family size. Let $\mathbf{Z}_n$ be the number of individuals in generation n , started from a single progenitor, $\mathbf{Z}_0 = 1$. Conditioning on the previous generation and using the tower property,

$$\mathbb{E}(\mathbf{Z}_{n+1}) = \mathbb{E}(\mathbb{E}(\mathbf{Z}_{n+1} \mid \mathbf{Z}_n)) = \mathbb{E}(\mu \mathbf{Z}_n) = \mu \mathbb{E}(\mathbf{Z}_n),$$

so $\mathbb{E}(\mathbf{Z}_n) = \mu^n$. When $\mu < 1$ the expected population decays geometrically and $\Pr\{\mathbf{Z}_n = 0\} \rightarrow 1$: extinction is certain. The mean alone settles the question in that regime, and nothing else in this chapter would be needed if the mean were all one wanted.

Generating functions supply the rest. Write $\phi(z) = \mathbb{E}(z^\xi)$ for the offspring probability generating function and $u_n = \Pr\{\mathbf{Z}_n = 0\}$ for the probability of extinction by generation n . Generation n is the union of the families produced by the individuals of generation $n - 1$, and those families are independent and identically distributed, so (11) gives $\mathbb{E}(z^{\mathbf{Z}_n} \mid \mathbf{Z}_{n-1} = m) = \phi(z)^m$ and hence $\phi_n(z) = \mathbb{E}(\phi(z)^{\mathbf{Z}_{n-1}}) = \phi_{n-1}(\phi(z))$. Iterating, the probability generating function of $\mathbf{Z}_n$ is the n -fold composition

$$\phi_n(z) = \mathbb{E}(z^{\mathbf{Z}_n}) = \underbrace{\phi(\phi(\cdots \phi(z)))}_{n \text{ times}}, \quad (20)$$

and evaluating at $z = 0$ gives the iteration on which almost everything below depends,

$$u_{n+1} = \phi(u_n), \quad u_0 = 0. \quad (21)$$

Extinction of a branching process is a fixed-point problem for a one-dimensional map.

3.1 Death or division

From here on we specialise to the binary — or *simple* — case, in which an individual either divides into two or dies:

$$\begin{aligned} p_2 &= p && \text{(division),} \\ p_0 &= 1 - p && \text{(death),} \\ p_i &= 0 && \forall i \neq 0, 2. \end{aligned}$$

The mean family size is then $\mu = 2p$, and the expected population descended from a single ancestor is

$$\mathbb{E}(\mathbf{Z}_n) = (2p)^n. \quad (22)$$

The process is *subcritical* when $2p < 1$, *critical* when $p = \frac{1}{2}$, and *supercritical* when $2p > 1$. Figure 1 shows realisations on either side of the critical point, drawn as genealogies.

The offspring probability generating function is a quadratic,

$$\phi(z) = \mathbb{E}(z^{\mathbf{Z}_1}) = pz^2 + (1 - p), \quad (23)$$

so that (21) reads

$$u_{n+1} = pu_n^2 + 1 - p. \quad (24)$$

The recursion admits a direct reading, which is worth having because it is the reading that generalises. If the founder dies at its first step, with probability $1 - p$, then extinction has already occurred. If instead it divides, with probability p , then extinction by generation $n + 1$ requires each of the two fresh lineages to die out within its own n generations, and those lineages are independent, so that event has probability u_n^2 .

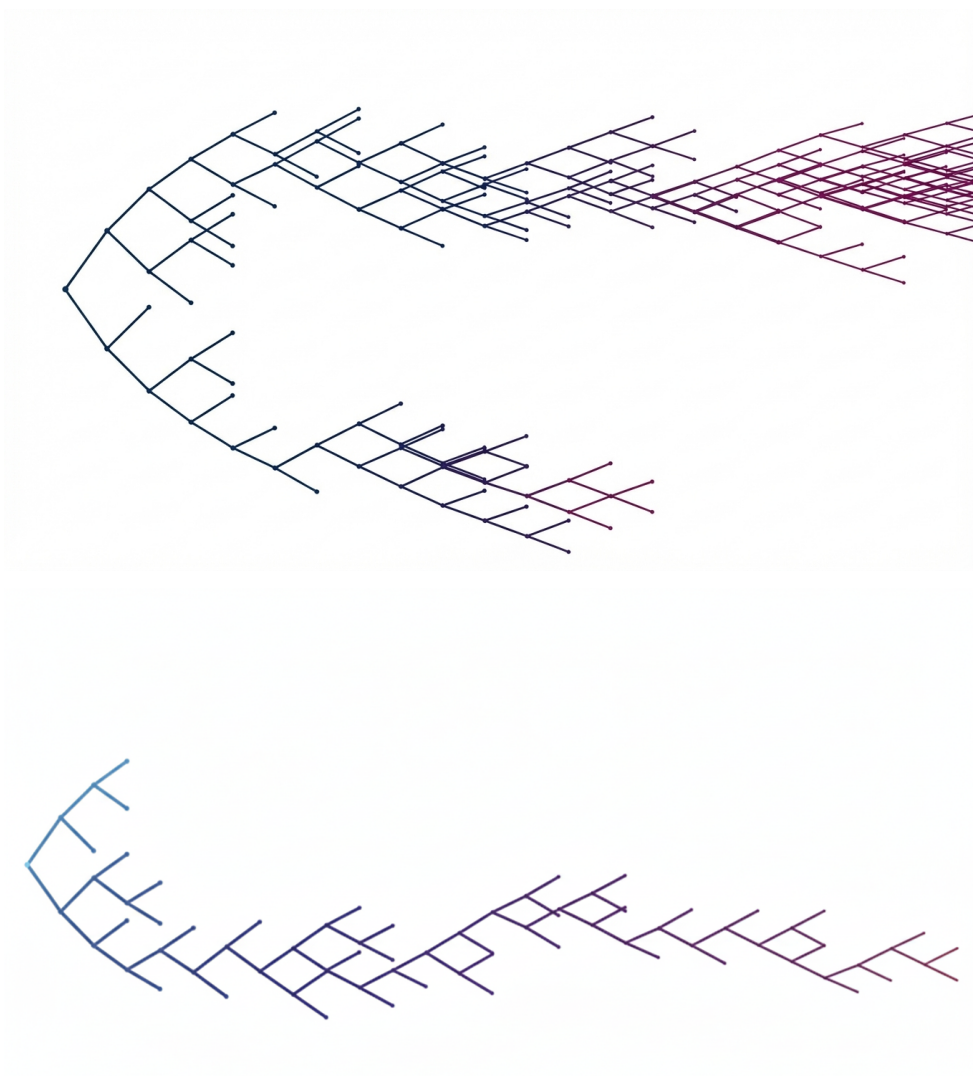


Figure 1: Realisations of the simple Galton–Watson process, drawn as genealogies with generation increasing to the right and colour encoding generation number. Top: supercritical, $p = 0.58$, in which the number of lines grows without bound. Bottom: subcritical, $p = 0.47$, in which the identical construction terminates. The two panels differ only in the value of p .

It is more convenient to track the complement,

$$S_n = 1 - u_n = \Pr \{ \mathbf{Z}_n > 0 \},$$

the probability that the process is still alive at generation n . Substituting $u_n = 1 - S_n$ into (24) and simplifying,

$$S_{n+1} = 2pS_n - pS_n^2, \quad S_0 = 1. \quad (25)$$

A population cannot return from extinction, so u_n is non-decreasing and bounded above by 1; equivalently S_n is non-increasing and bounded below by 0, and therefore converges to some $S_\infty \in [0, 1]$. The recursion (25) is the central object of [sections 4 to 6](#), and every difficulty encountered there is a difficulty about this one quadratic map.

3.2 Certainty of extinction

Since S_n converges, its limit must be a fixed point of (25). Setting $S_{n+1} = S_n = S_\infty$,

$$S_\infty = 2pS_\infty - pS_\infty^2 \iff pS_\infty \left(S_\infty + \frac{1}{p} - 2 \right) = 0,$$

with roots

$$S_\infty = 0 \quad \text{and} \quad S_\infty = 2 - \frac{1}{p} = \frac{2p-1}{p}.$$

For $p < \frac{1}{2}$ the second root is negative and carries no probabilistic meaning, so the only admissible limit is $S_\infty = 0$ and extinction is certain. At $p = \frac{1}{2}$ the two roots collide at the origin and extinction remains certain, now marginally. For $p > \frac{1}{2}$ a positive root appears at $S_\infty = (2p-1)/p$, and it is this root that is attained: the process survives forever with positive probability. Figure 2 displays the transition as a change in the stability of the origin.

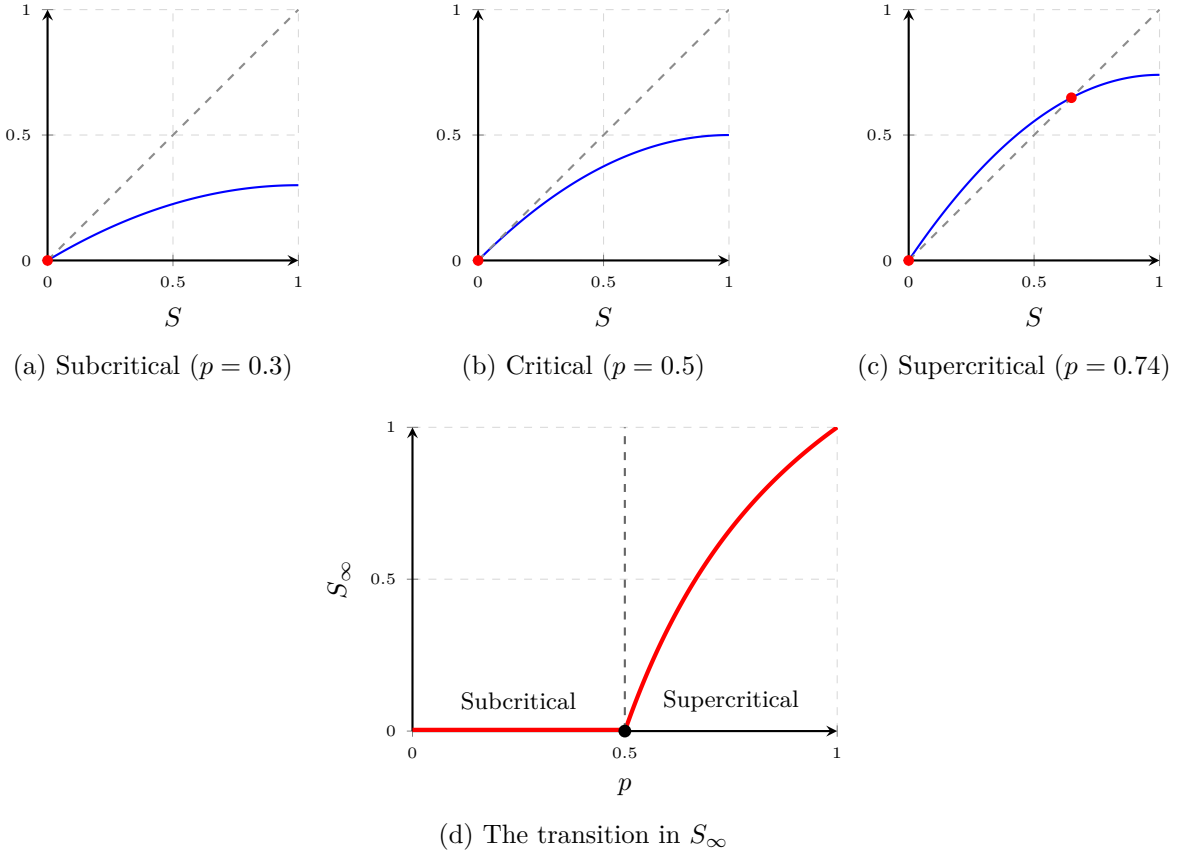


Figure 2: Steady states of the survival recursion (25). Panels (a)–(c) plot the map $S \mapsto 2pS - pS^2$ in blue against the diagonal, dashed; fixed points are the intersections, marked in red. Below and at criticality the origin is the only fixed point in $[0, 1]$ and it attracts, so extinction is certain. At $p = \frac{1}{2}$ the map is tangent to the diagonal at the origin. For $p > \frac{1}{2}$ the origin turns repelling and a stable positive fixed point $S_\infty = (2p-1)/p$ separates from it, traced in panel (d). The survival probability is identically zero below $p = \frac{1}{2}$ and rises continuously above it.

3.3 Criticality: a power law and an infinite expected lifetime

The critical case sits awkwardly between the two others and deserves separate treatment. At $p = \frac{1}{2}$ the expected population is constant, $\mathbb{E}(\mathbf{Z}_n) = 1^n = 1$ for every n , and yet extinction is certain. The two facts are reconciled by the expected *lifetime* being infinite: rare realisations survive for enormous times and grow enormously large, and they do so at exactly the rate needed to hold the mean at one. Demonstrating this is worthwhile in itself, and the same calculation supplies the asymptotics of S_n that [section 6](#) will need.

Let

$$T = \inf\{n : \mathbf{Z}_n = 0\}$$

be the extinction time, so that the process passes through generations $0, 1, \dots, T-1$ and is dead at T . Writing T as a sum of indicators and taking expectations term by term — legitimate because every term is non-negative —

$$T = \sum_{n=0}^{\infty} \mathbf{1}_{\{T > n\}}, \quad \mathbb{E}(T) = \sum_{n=0}^{\infty} \Pr\{T > n\} = \sum_{n=0}^{\infty} \Pr\{\mathbf{Z}_n > 0\} = \sum_{n=0}^{\infty} S_n. \quad (26)$$

The question is therefore how fast S_n decays at criticality.

Setting $p = \frac{1}{2}$ in [\(25\)](#) removes the linear term entirely and leaves

$$S_{n+1} - S_n = -\frac{1}{2}S_n^2.$$

Once n is large the unit generational increment is small compared with the scale on which S_n varies, so the difference on the left is well approximated by a derivative:

$$\frac{dS}{dn} = -\frac{1}{2}S^2, \quad \text{with general solution} \quad S(n) = \frac{2}{n + \text{const}}.$$

The approximation is heuristic — it is the continuum limit of a difference equation, and its constant of integration is not determined by it — but the decay rate it predicts is correct:

$$S_n \sim \frac{2}{n} \quad (n \rightarrow \infty), \quad (27)$$

as the simulations of [figure 4\(a\)](#) confirm. Substituting into [\(26\)](#), the tail of the sum behaves like twice the harmonic series,

$$\mathbb{E}(T) = \sum_{n=0}^{\infty} S_n \sim \sum_{n=1}^{\infty} \frac{2}{n},$$

which diverges ([figure 3](#)). Since every S_n is positive, $\mathbb{E}(T) = \infty$: the expected lifetime of the critical Galton–Watson process is infinite even though extinction occurs with probability one.

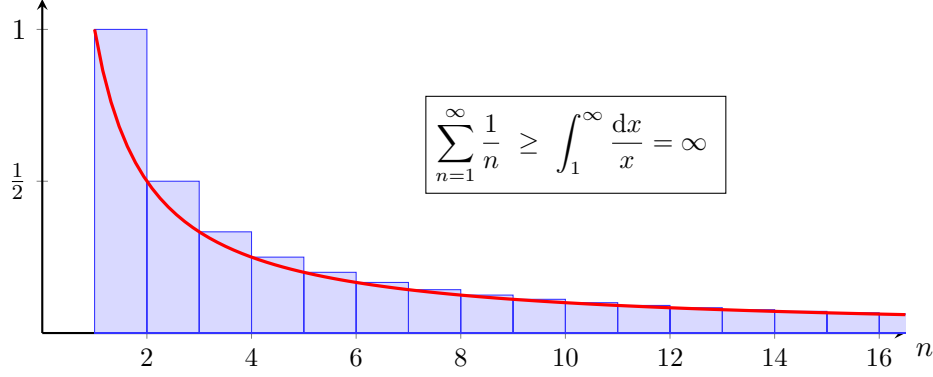


Figure 3: Divergence of the harmonic series. Each blue rectangle has area $1/n$ and contains the region under the red curve $1/x$ on $[n, n+1]$, so the areas sum to more than $\int_1^{\infty} x^{-1} dx = \infty$. Because $S_n \sim 2/n$ at criticality, the same divergence carries over to $\mathbb{E}(T) = \sum_n S_n$ in (26).

The heavy tail responsible for the divergence is visible directly in simulation. A power-law distributed quantity carries a modest but non-negligible chance of an extraordinarily large realisation, and a sample mean computed from such a quantity does not settle: the larger the sample, the larger the mean, because the occasional enormous value keeps pushing the normalised sum upward. Figure 4 collects the three classical critical exponents, of which (27) is the first.

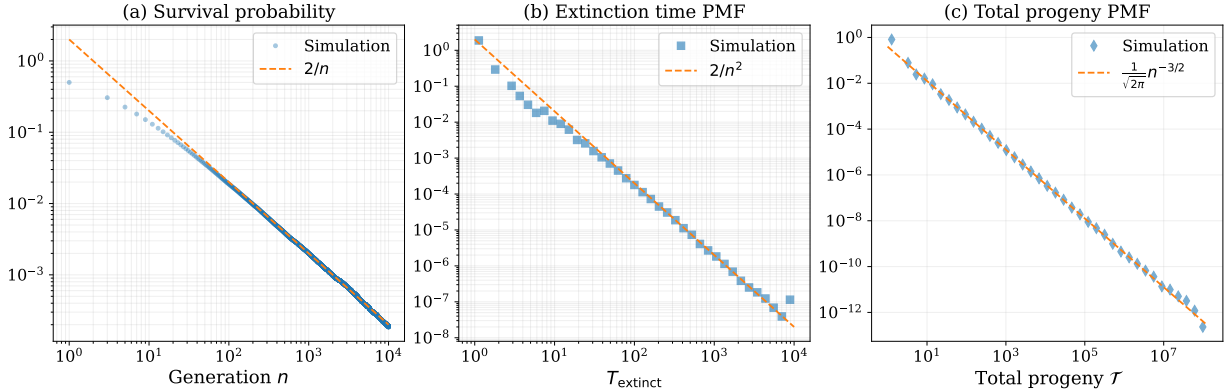


Figure 4: Power-law behaviour at criticality, $p = 0.5$, from 10^6 simulated realisations. (a) Survival probability $S_n = \Pr\{\mathbf{Z}_n > 0\}$, showing the late-generation scaling $S_n \sim 2/n$ of (27). (b) Probability mass function of the extinction time $T = \inf\{n : \mathbf{Z}_n = 0\}$, with heavy tail $\Pr\{T = n\} \sim 2/n^2$. (c) Probability mass function of the total progeny $\mathcal{T} = \sum_{k \geq 0} \mathbf{Z}_k$, displaying the classical critical branching asymptotic $\Pr\{\mathcal{T} = n\} \sim n^{-3/2}$.

Criticality is thus the boundary at which the mean stops being informative in either direction. Below it the population is doomed and the mean says so; above it the population may establish itself and the mean says roughly how fast. At $p = \frac{1}{2}$ the mean is one forever while the process is certain to die, and the only way to recover a useful statement is to condition on survival. That is the subject of the next section.

4 Conditioning on survival

Absorption is often inevitable, as it is here for $p < \frac{1}{2}$, and yet long before it occurs a process may settle into a distribution that no longer changes with time — a distribution of the population *given* that absorption has not yet happened. This is *quasi-stationarity*, and the limit it describes is the quasi-stationary, or Yaglom, distribution [4, 2]. This section establishes it for the binary Galton–Watson process by computing the limiting conditional mean and variance, and in doing so it produces the constant that the rest of the chapter is about.

4.1 The survival probability at late times

Everything follows from the decay of S_n . Take $p < \frac{1}{2}$, so that death of an individual is more probable than its division. Then $S_n \rightarrow 0$, and as it does so the quadratic term in

$$S_{n+1} = 2pS_n - pS_n^2$$

becomes negligible beside the linear one. The recursion is therefore increasingly well approximated by the linear map

$$S_{n+1} \approx 2p S_n, \quad (28)$$

whose solution is a pure geometric. The late-time behaviour of the survival probability is accordingly

$$S_n \sim A(p) (2p)^n \quad (n \rightarrow \infty), \quad A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n}, \quad (29)$$

and the whole of the nonlinearity has been pushed into the single constant $A(p)$. That constant records the cumulative effect of the quadratic term over the early generations, before the linear regime of (28) takes hold; it is what the process remembers of its own beginning. The limit exists and is finite and strictly positive for $p \in [0, \frac{1}{2})$, a fact proved by an infinite product in [section 6.1](#) and sharpened to $A(p) \in (0, \frac{1}{2}]$ in [section 6.2](#).

Remark 4.1. (29) is an asymptotic equivalence, not an identity. For $p < \frac{1}{2}$ the limit of S_n is 0, and one must not write $\lim_n S_n = A(p)(2p)^n$, whose right-hand side still depends on n . What converges is the ratio $S_n/(2p)^n$.

4.2 The conditional mean

Split the expected population size into contributions from surviving and extinct realisations:

$$\mathbb{E}(\mathbf{Z}_n) = \Pr\{\mathbf{Z}_n = 0\} \mathbb{E}(\mathbf{Z}_n | \mathbf{Z}_n = 0) + \Pr\{\mathbf{Z}_n > 0\} \mathbb{E}(\mathbf{Z}_n | \mathbf{Z}_n > 0).$$

The first term vanishes identically, so with $\Pr\{\mathbf{Z}_n > 0\} = S_n$ and $\mathbb{E}(\mathbf{Z}_n) = (2p)^n$ from (22),

$$\mathbb{E}(\mathbf{Z}_n | \mathbf{Z}_n > 0) = \frac{\mathbb{E}(\mathbf{Z}_n)}{S_n} = \frac{(2p)^n}{S_n}. \quad (30)$$

The numerator is known exactly and the content sits entirely in the denominator, which (29) supplies:

$$\mathbb{E}(\mathbf{Z}_n | \mathbf{Z}_n > 0) \xrightarrow{n \rightarrow \infty} \frac{(2p)^n}{A(p)(2p)^n} = \frac{1}{A(p)}. \quad (31)$$

The geometric factors cancel exactly. At late times the surviving population therefore has a constant expected size, and that size is the reciprocal of $A(p)$; for this reason $1/A(p)$ is called the *quasi-stationary mean* below. Two exponentially small quantities have been divided into one another to leave something of order unity, which is what conditioning on a vanishing event generally does and why the unconditional mean was so uninformative.

Figure 5 shows the convergence, and also shows the obstacle that section 6 will have to deal with: the closer p lies to $\frac{1}{2}$, the higher the plateau and the more generations are needed to reach it.

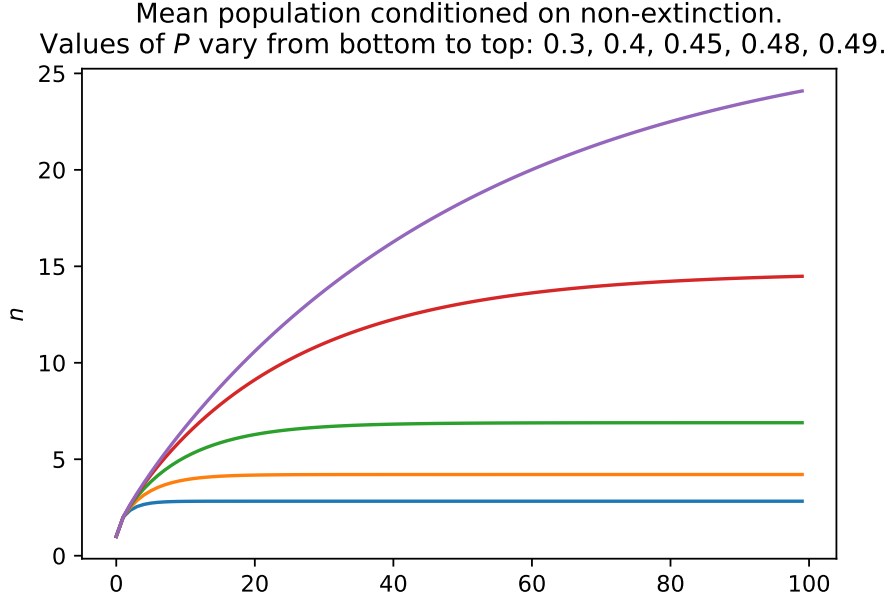


Figure 5: The conditional mean $(2p)^n/S_n$ against generation number n , with S_n obtained by iterating (25). From bottom to top, $p = 0.3, 0.4, 0.45, 0.48, 0.49$. Each curve approaches the constant $1/A(p)$, whose values are 2.825, 4.208, 6.893, 14.703 and 27.476, computed as in section 6.7. The lowest three have essentially converged by $n = 100$; the $p = 0.49$ curve has reached only 24.1 of its limiting 27.5, which is an early sign of the near-critical slowdown quantified in table 1. (The vertical axis is mislabelled n in the original plotting script; the quantity plotted is the conditional mean.)

4.3 The conditional variance

A stable mean is not yet quasi-stationarity. To show that the conditional *distribution* settles we need at least the second moment, and obtaining it is a useful exercise in the composition structure of (20).

4.3.1 The second moment

Work with the moment generating function $M_n(t) = \mathbb{E}(e^{tZ_n})$, which is the probability generating function evaluated on the exponential,

$$M_n(t) = \phi_n(e^t), \quad \mathbb{E}(Z_n^k) = M_n^{(k)}(0), \quad (32)$$

and note the two facts that make the calculation short. First, $M_n(0) = 1$ for every n . Second, the offspring probability generating function (23) has the linear derivative $\phi'(z) = 2pz$. Since

$\phi_n = \phi \circ \phi_{n-1}$, the chain rule gives

$$M'_n(t) = \phi'(\phi_{n-1}(e^t)) M'_{n-1}(t) = 2p M_{n-1}(t) M'_{n-1}(t), \quad (33)$$

a recursion in which the derivative of the composition never has to be expanded. Evaluating (33) at $t = 0$ recovers the mean, $M'_n(0) = 2p M'_{n-1}(0)$ and hence $M'_n(0) = (2p)^n$, as it must by (22). Differentiating (33) once more,

$$M''_n(t) = 2p \left((M'_{n-1}(t))^2 + M_{n-1}(t) M''_{n-1}(t) \right),$$

and setting $t = 0$ turns this into a linear recursion for the second moments. Writing $m_n = \mathbb{E}(\mathbf{Z}_n^2) = M''_n(0)$ and $\mu = 2p$,

$$m_n = \mu \mu^{2(n-1)} + \mu m_{n-1} = \mu^{2n-1} + \mu m_{n-1}, \quad m_0 = 1. \quad (34)$$

Unrolling (34) gives $m_n = \mu^n + \sum_{j=1}^n \mu^{n-j} \mu^{2j-1} = \mu^n (1 + \sum_{i=0}^{n-1} \mu^i)$, and summing the finite geometric series,

$$\mathbb{E}(\mathbf{Z}_n^2) = (2p)^n \left[1 + \frac{1 - (2p)^n}{1 - 2p} \right] = (2p)^n \left(1 + \frac{1}{1 - 2p} \right) - (2p)^{2n} \left(\frac{1}{1 - 2p} \right). \quad (35)$$

At $n = 1$ the right-hand side returns $2p + (2p - 4p^2)/(1 - 2p) = 4p$, which is $\mathbb{E}(\mathbf{Z}_1^2)$ for the binary offspring law, so the signs are as they should be.

4.3.2 Late-time behaviour

Subtracting $\mathbb{E}(\mathbf{Z}_n)^2 = (2p)^{2n}$ from (35) gives a variance that factorises neatly,

$$\text{Var}(\mathbf{Z}_n) = \left(1 + \frac{1}{1 - 2p} \right) \left((2p)^n - (2p)^{2n} \right), \quad (36)$$

so for $2p < 1$ the variance decays geometrically like $(2p)^n$, at the same rate as the mean. It is tempting to read stationarity of the conditional variance directly off (36), but that would be too quick: the conditional variance is not the unconditional one divided by anything simple. The safe route is through the conditional moments themselves. Dividing by S_n and using (29), with the $(2p)^{2n}$ terms negligible beside $(2p)^n$,

$$\begin{aligned} \mathbb{E}(\mathbf{Z}_n^2 \mid \mathbf{Z}_n > 0) &= \frac{\mathbb{E}(\mathbf{Z}_n^2)}{S_n} \xrightarrow{n \rightarrow \infty} \frac{1}{A} \left(1 + \frac{1}{1 - 2p} \right), \\ \mathbb{E}(\mathbf{Z}_n \mid \mathbf{Z}_n > 0) &= \frac{\mathbb{E}(\mathbf{Z}_n)}{S_n} \xrightarrow{n \rightarrow \infty} \frac{1}{A}. \end{aligned}$$

Both limits are finite, so the conditional variance tends to a constant,

$$\text{Var}(\mathbf{Z}_n \mid \mathbf{Z}_n > 0) \longrightarrow \frac{1}{A} \left(1 + \frac{1}{1 - 2p} \right) - \frac{1}{A^2}. \quad (37)$$

The first two moments of the conditional population therefore both stabilise, and the same argument carried to higher moments gives the full quasi-stationary distribution. As a check on (37), let $p \rightarrow 0$: there $A \rightarrow \frac{1}{2}$ and the bracket tends to 2, so the limiting conditional variance is $2 \cdot 2 - 4 = 0$. Deep in the subcritical regime a surviving population consists of exactly two individuals and has no spread at all, which is the parity effect made precise in [proposition 6.3](#).

What matters for the rest of the chapter is that (31) and (37) are both expressed in terms of p and the single unknown constant $A(p)$. Everything now turns on that constant.

4.4 A continuum view of the survival recursion

Before attacking $A(p)$ directly it is worth recording a continuum approximation to (25), both because it recurs later and because its failure is instructive. Write $\varepsilon = 1 - 2p$ for the distance from criticality, so that the exact difference is

$$S_{n+1} - S_n = -\varepsilon S_n - pS_n^2. \quad (38)$$

For large n the unit increment is small on the scale over which S varies, and replacing the left-hand side by a derivative gives a Riccati equation,

$$\frac{dS}{dn} = -\varepsilon S - pS^2. \quad (39)$$

The substitution $u = 1/S$ linearises it to $u' = \varepsilon u + p$, with solution $u(n) = Ke^{\varepsilon n} - p/\varepsilon$ and hence

$$S(n) = \frac{\varepsilon}{K\varepsilon e^{\varepsilon n} - p}, \quad (40)$$

where K is a constant of integration fixed by the initial condition. Both known regimes are contained in (40). Setting $\varepsilon = 0$ recovers the critical scaling $S \sim 2/n$ of (27); keeping $\varepsilon > 0$ recovers the geometric decay $S \sim A(2p)^n$ of (29), since $(2p)^n = e^{n \ln(1-\varepsilon)}$ agrees with $e^{-\varepsilon n}$ to leading order in ε .

What (40) does not contain is K . The continuum approximation is valid only once n is large, whereas K is determined by the early generations in which it fails, and K is — up to the identification just made — precisely $A(p)$. No argument of this kind will produce it. Section 6.3 shows how much can nevertheless be extracted from (39) in the near-critical limit, where the early generations occupy a vanishing fraction of the process's life.

5 Survival of small populations

Quasi-stationarity describes what a surviving population looks like once it has survived for a long time. The complementary question is how it gets there from a small founding number, and near criticality the answer is dominated by the first few generations. This section works through that early regime, records the discrete birth–death–catastrophe calculation that later chapters build on, and then computes the continuous-time counterpart of $A(p)$, where a closed form is available and the comparison is informative.

5.1 The first few generations

Take the binary process near criticality, so that division and death each have probability close to a half. A founding lineage then dies at its very first round almost half the time. Should it pass that round, its two offspring must each fail at their own first attempt for the lineage to end there, which happens with probability about 0.25; so the lineage reaches the second generation with probability roughly

$$0.375 \approx \underbrace{(1 - 0.5)}_{\text{no extinction at the first step}} \times \underbrace{(1 - 0.5 \times 0.5)}_{\text{nor at the second}}.$$

Iterating (24) and (25) from $u_0 = 0$, $S_0 = 1$ makes this exact. At precise criticality the first four values are

$$u_n = 0.500, 0.625, 0.695, 0.741 \quad (3 \text{ s.f., } n = 1, 2, 3, 4),$$

so survival decays quickly over the first few rounds of replication, and it does so almost as fast just above criticality as just below (figure 6). Whatever advantage supercriticality confers, it is not an

early-generation advantage: the curves for $p = 0.45, 0.5$ and 0.55 have barely separated by $n = 4$, and nearly all of the extinction risk has already been spent.

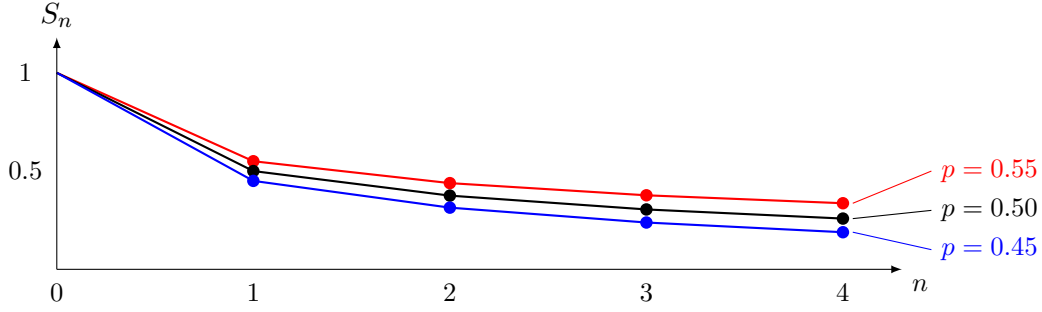


Figure 6: Survival probability over the first four generations from a single progenitor, obtained by iterating (25). The decay is steep even in the supercritical case $p = 0.55$, and the three curves remain close together throughout. The difference between sub- and supercriticality is a difference in late behaviour, not in early risk.

5.2 An initial cohort of size k

Now let the process begin with $\mathbf{Z}_0 = k$ individuals rather than one — a group small to moderate in number, perhaps $k = 10$. Under supercritical branching, growth becomes more predictably exponential as the count rises, because a large population lets the expected dynamics show through the noise of individual events. A founding cohort is precisely the regime in which that noise still dominates.

The persistence of such a cohort is a question about its lineages. All k founders branch independently, so the population is extinct at generation n exactly when every one of the k lineages has died out by then, each with probability u_n . Hence

$$S_n^{(k)} = 1 - (u_n)^k, \quad (41)$$

which is (11) in its most elementary form. Figures 7 and 8 plot $S_n^{(k)}$ out to 500 generations for $k = 1, 10, 100, 1000$.

The effect visible in figure 7 — that a larger first cohort gives improved longevity — is thus strongly modulated by criticality. Persistence improves rapidly as p approaches $\frac{1}{2}$ from below, and continues to improve dramatically into supercriticality, where a large k converts a modest per-lineage survival probability into near-certain establishment. What a founding cohort buys, in other words, depends almost entirely on a parameter it does not itself control.

5.3 An aside on early replication

The same conditioning effect has been identified in a setting far from the present one: the longevity and diversity of phylogenetic clades. Clades that have persisted through the ages, and which are often the more diverse, turn out to have experienced an above-average early rate of species origination — an observation drawn from the fossil record itself. Budd and Mann [5] argue that this empirical pattern may be a statistical consequence rather than a biological one. Run a large number of homogeneous birth–death processes, retain only those that happen to survive to some arbitrary large time — “the present” — and the retained set is automatically enriched for early

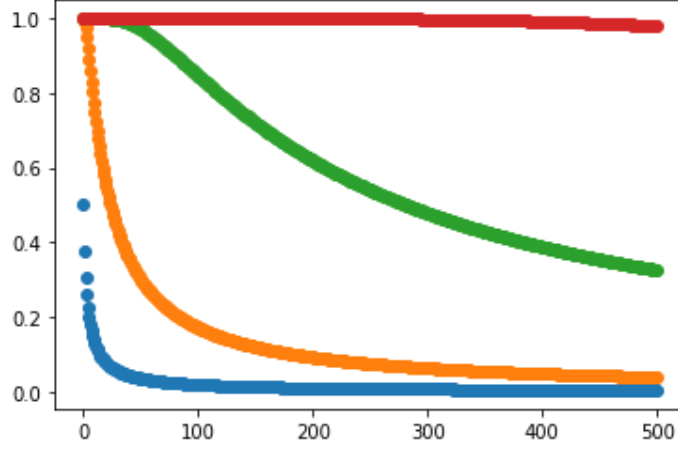


Figure 7: Survival probability $S_n^{(k)}$ from (41) out to 500 generations at exact criticality, $p = 0.5$, for $k = 1, 10, 100, 1000$ (blue through to red). A larger founding cohort buys longevity, but at criticality every curve is still headed for zero.

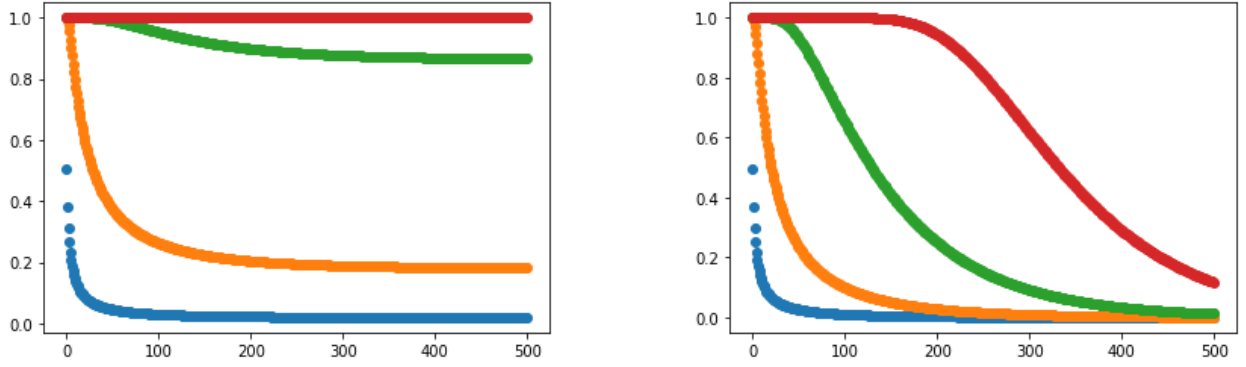


Figure 8: As figure 7, but with p displaced slightly from criticality: $p = 0.505$ on the left (supercritical) and $p = 0.495$ on the right (subcritical). The benefit of a large founding cohort is acutely sensitive to which side of $\frac{1}{2}$ the process sits on. A displacement of half a per cent separates curves that were nearly coincident in figure 7.

rapid origination, with no change in the underlying rates required. The apparent slowdown in diversification that follows is then not a slowdown at all but a reversion to the mean, an effect known as the *push of the past*.

That is the conditioning of section 4 seen from the other side. There we conditioned a subcritical process on survival and found its mean settling at $1/A(p)$ instead of decaying; here one conditions a critical or supercritical process on survival to the present and finds its early growth inflated. In both cases the conditioning, not the mechanism, does the work, and in both cases the effect is largest where the survival probability is smallest.

The parallel suggests a modelling hypothesis worth carrying forward: that early replication may strongly affect eventual survival probabilities even when later per-capita rates are unchanged. Figure 6 is the reason to take it seriously, since the first few generations are where nearly all of the extinction risk is concentrated.

5.4 Discrete birth–death–catastrophe

Later chapters replace the pure birth–death mechanism by one in which the whole population can be eliminated at a stroke. It is worth recording here what that addition does in discrete time, since the answer is partly inherited from the Galton–Watson analysis above.

Up to the moment of catastrophe, a discrete-time birth–death–catastrophe process behaves exactly as the basic Galton–Watson model as far as its limiting conditional distribution is concerned. Conditioning on non-rupture, and conditioning also on non-extinction by death, the process exhibits a stationary limiting distribution of the kind derived in [section 4](#). Whether a quasi-stationary distribution survives when one conditions on *neither* death nor catastrophe is a separate question, and one best settled by simulation. Either way it is useful to have the probability of non-catastrophe in hand.

Rupture at step n

Let κ be the probability that a given individual initiates catastrophe at a given step. A single individual fails to cause rupture with probability $1 - \kappa$, two with $(1 - \kappa)^2$, and a cohort of size N with $(1 - \kappa)^N$.

Take first the case of birth and catastrophe only, with no death. The census is then known with certainty at every generation: from a single progenitor the population simply doubles at each step prior to catastrophe. The process therefore comes to its abrupt halt in generation n with probability

$$1 - (1 - \kappa)^{2^{n-1}}, \quad (42)$$

one minus the probability that none of the 2^{n-1} individuals present caused it.

Survival of the process

Along the same lines, and still without death, let S_n be the probability that the process survives to generation n . Survival to generation 0 is certain, $S_0 = 1$. Survival to generation 1 requires only that the founder replicated without rupture, $S_1 = 1 - \kappa$. Persistence to step 2 requires that the founder divided and that each of its two offspring did likewise, so $S_2 = (1 - \kappa)(1 - \kappa)^2$. Every prior step must be passed, and the exponents form a geometric series:

$$S_n = \prod_{i=0}^{n-1} (1 - \kappa)^{2^i} = (1 - \kappa)^{\sum_{i=0}^{n-1} 2^i} = (1 - \kappa)^{2^n - 1}. \quad (43)$$

Survival therefore fails doubly exponentially in n , which is a much more severe constraint than the geometric decay of [\(29\)](#): with catastrophe present the population’s own growth is what destroys it.

Death included

The difficulty appears as soon as death is allowed, because the size of the living population is then no longer known with certainty from one generation to the next. The natural guess is that the probabilities somehow shake out when the expected population is used in place of the actual one,

$$S_n \stackrel{?}{=} (1 - \kappa)^{\sum_{i=0}^{n-1} \mathbb{E}(\mathbf{Z}_i)}, \quad (44)$$

and it is worth being explicit about why [\(44\)](#) is almost certainly not an identity. The map $N \mapsto (1 - \kappa)^N$ is convex, so Jensen’s inequality gives $\mathbb{E}((1 - \kappa)^{\mathbf{Z}_i}) \geq (1 - \kappa)^{\mathbb{E}(\mathbf{Z}_i)}$: replacing the

random census by its mean systematically underestimates the chance of getting through a step. Nor are the steps independent, since surviving one is evidence of a smaller population, which raises the chance of surviving the next. Both effects push the same way, so (44) is better read as a lower bound than as an equality. How tight a bound it is remains a numerical question, and the exact treatment — through the rates, the master equation and its generating function, as in [section 2](#) — is developed in later chapters for birth–death–catastrophe processes in continuous time.

5.5 Continuous-time birth–death

The discrete process resists a closed form for $A(p)$, as [section 6](#) shows at length. Its continuous-time counterpart does not, and the comparison between the two turns out to be informative, so we record the continuous-time constant here.

Everything needed is already in [section 2.4](#). For the linear birth–death process with per-capita rates λ and μ started from N individuals, the extinction probability is $p_0(t)$ of (19) and the survival probability is $S^{(N)}(t) = 1 - p_0(t)$. Exactly as in [section 4.2](#), conditioning on survival gives

$$\mathbb{E}(\mathbf{X}_t \mid \mathbf{X}_t > 0) = \frac{\mathbb{E}(\mathbf{X}_t)}{S^{(N)}(t)}, \quad (45)$$

and the late-time limit follows for any N on applying L'Hôpital's rule. It is quicker to take $N = 1$ and compute directly. With $\mathbb{E}(\mathbf{X}_t) = e^{(\lambda-\mu)t}$ from (18) and

$$S(t) = 1 - \frac{\mu - \mu e^{(\mu-\lambda)t}}{\lambda - \mu e^{(\mu-\lambda)t}} = \frac{\lambda - \mu}{\lambda - \mu e^{(\mu-\lambda)t}},$$

equation (45) gives

$$\mathbb{E}(\mathbf{X}_t \mid \mathbf{X}_t > 0) = \frac{\lambda e^{(\lambda-\mu)t} - \mu}{\lambda - \mu}.$$

For $\mu > \lambda$ the exponential decays and the limiting conditional mean is a constant,

$$\lim_{t \rightarrow \infty} \mathbb{E}(\mathbf{X}_t \mid \mathbf{X}_t > 0) = \frac{\mu}{\mu - \lambda} = \frac{1}{1 - \lambda/\mu} \left(= \frac{1}{A_c} \right). \quad (46)$$

So the subcritical birth–death process is quasi-stationary in the same sense as the subcritical Galton–Watson process, and with the same structure: a conditional mean that is the reciprocal of a constant. To compare the two we need a common parameter. An individual's next event is a division with probability $p = \lambda/(\lambda + \mu)$ by the race (5), so that $\lambda/\mu = p/(1 - p)$ and

$$\frac{1}{A_c} = \frac{1 - p}{1 - 2p}, \quad \text{that is} \quad A_c(p) = \frac{1 - 2p}{1 - p}. \quad (47)$$

The subscript marks this as the continuous-time constant, to be distinguished throughout from the discrete-time $A(p)$ of (29).

[Figure 9](#) plots $A_c(p)$ against numerically computed values of $A(p)$ and raises an obvious question. Why does the continuous curve start at 1 where the discrete one starts at $\frac{1}{2}$, and why does halving A_c not superpose them, given that at the other end of the range they meet? The answer needs machinery that [section 6](#) has still to build; it is given in [section 6.4](#), and it turns out that $A_c(p)/2$ is an exact lower bound for $A(p)$, attained only in the limit $p \rightarrow 0$.



Figure 9: The continuous-time constant $A_c(p) = (1 - 2p)/(1 - p)$ (green) against high-precision numerical values of the discrete-time constant $A(p)$ (red points). Both vanish at $p = \frac{1}{2}$, but at $p = 0$ the continuous curve starts at 1 and the discrete one at $\frac{1}{2}$, and no constant factor relates them in between. The black line is a formula obtained by symbolic regression ([section 6.5](#)); it tracks the points closely but is not exact. [Section 6.4](#) accounts for the discrepancy exactly.

6 The constant $A(p)$

Both quasi-stationary results of [section 4](#) reduced to the single constant

$$A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n}, \quad p \in [0, \tfrac{1}{2}), \quad (48)$$

whose reciprocal is the quasi-stationary mean. Approximate values are easy to obtain: iterate (25) until S_n is very nearly stationary, then form $S_n/(2p)^n$. Repeating over a range of p traces the curve of [figure 9](#), and the procedure works well over most of the interval. It fails as $p \rightarrow \frac{1}{2}^-$, because the number of generations needed for S_n to reach near-stationarity grows without bound.

No formula in elementary functions is known for $A(p)$, and a good deal of effort went into looking for one. This section sets out what was found. [Section 6.1](#) gives an exact infinite product; [section 6.2](#) converts it into an exact series with two-sided bounds that pin $A(p)$ between the continuous-time constant of [section 5.5](#) and twice it; [section 6.3](#) extracts the near-critical asymptotic $A(p) \sim 2(1 - 2p)$; [section 6.4](#) settles the discrepancy between the discrete and continuous curves; [section 6.5](#) records the symbolic-regression searches; and [section 6.6](#) gives the dynamical argument that explains why those searches were doomed. [Section 6.7](#) draws the practical conclusion.

6.1 An infinite-product representation

Begin by putting the survival recursion into its standard form. Substituting $S_n = 2w_n$ into (25) gives

$$w_{n+1} = 2p w_n (1 - w_n) = r w_n (1 - w_n), \quad r = 2p, \quad w_0 = \tfrac{1}{2}, \quad (49)$$

which is the logistic map — the same map that exhibits the period-doubling route to chaos as r increases from 0 to 4 (figure 10). The subcritical branching window is $r \in [0, 1)$, at the extreme left of that picture, where the origin is a globally attracting fixed point and the dynamics are as simple as they can be. That this innocuous corner of the logistic family is nevertheless enough to obstruct a closed form is the substance of section 6.6.

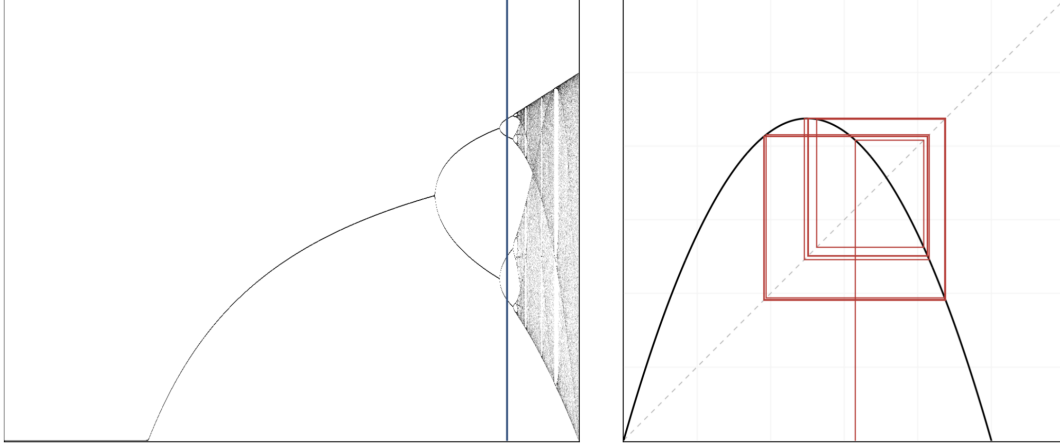


Figure 10: Bifurcation diagram of the logistic map $w \mapsto rw(1-w)$, showing the period doubling of stable states and the onset of chaos. The blue sample line on the left is at $r = 3.5255$, and the right-hand plot shows a trajectory of the map at that same r . The regime relevant to subcritical branching, $r = 2p \in [0, 1)$, lies off the left-hand edge of this diagram.

In terms of w , the definition (48) reads

$$A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n} = 2 \lim_{n \rightarrow \infty} \frac{w_n}{r^n}. \quad (50)$$

Any sequence can be written as a telescoping product of its own ratios,

$$x_n = x_0 \prod_{k=0}^{n-1} \frac{x_{k+1}}{x_k}, \quad \text{so} \quad \lim_{n \rightarrow \infty} x_n = x_0 \prod_{k=0}^{\infty} \frac{x_{k+1}}{x_k},$$

and applied to $x_n = w_n/r^n$ the ratio simplifies completely:

$$\frac{x_{k+1}}{x_k} = \frac{w_{k+1}}{w_k} \cdot \frac{1}{r} = \frac{rw_k(1-w_k)}{w_k} \cdot \frac{1}{r} = 1 - w_k.$$

The logistic nonlinearity has been traded for a product of its own iterates. With $x_0 = w_0 = \frac{1}{2}$ and $A(p) = 2 \lim_n x_n$ we obtain $A(p) = \prod_{k \geq 0} (1 - w_k)$, and pulling out the factor $1 - w_0 = \frac{1}{2}$,

$$A(p) = \frac{1}{2} \prod_{k=1}^{\infty} (1 - w_k) = \frac{1}{2} \prod_{k=1}^{\infty} \left(1 - \frac{S_k}{2}\right). \quad (51)$$

Convergence

For $0 < w_k < 1$ the product $\prod_{k \geq 1} (1 - w_k)$ converges to a strictly positive limit if and only if $\sum_{k \geq 1} w_k < \infty$. From (49) with $0 < r < 1$ and $0 < w_n < 1$ we have $w_{n+1} < rw_n$, so by induction $w_n \leq w_0 r^n$, which is summable. The product therefore converges to a positive finite value. This also supplies what was asserted in section 4.1: the limit (48) defining $A(p)$ exists, and lies strictly between 0 and $\frac{1}{2}$ for $p \in (0, \frac{1}{2})$.

Is this an advance?

Only a modest one. (51) is exact and its partial products decrease monotonically, so truncation gives a rigorous upper bound on $A(p)$ at every order, which the raw ratio $S_n/(2p)^n$ does not. Computing the factors still requires iterating the nonlinear map to generate the S_k , however, so the near-critical difficulty is untouched. What the product does supply is the route to the sharper result of the next subsection.

6.2 An exact series and two-sided bounds

A product of terms close to one is a sum in disguise, and the sum turns out to be far more informative. Keep $r = 2p$ and note that the survival recursion factorises as

$$S_{n+1} = rS_n \left(1 - \frac{S_n}{2}\right).$$

Define

$$v_n = \frac{r^n}{S_n},$$

so that $v_n \rightarrow 1/A(p)$ by (48). The increments of v_n collapse:

$$v_{n+1} - v_n = \frac{r^{n+1}}{rS_n \left(1 - \frac{S_n}{2}\right)} - \frac{r^n}{S_n} = \frac{r^n}{S_n} \left[\frac{1}{1 - \frac{S_n}{2}} - 1 \right] = \frac{r^n}{S_n} \cdot \frac{S_n/2}{1 - \frac{S_n}{2}} = \frac{r^n}{2 - S_n},$$

and summing from $v_0 = 1/S_0 = 1$ gives an exact representation of the quasi-stationary mean.

Proposition 6.1 (Series representation). *For $p \in [0, \frac{1}{2})$, with S_n generated by (25),*

$$\frac{1}{A(p)} = 1 + \sum_{n=0}^{\infty} \frac{(2p)^n}{2 - S_n}. \quad (52)$$

Unlike the product, this representation isolates the geometric decay in a factor that does not depend on the dynamics at all, and confines the whole of the nonlinearity to the denominator $2 - S_n$. That is what makes it useful: S_n is confined to $(0, 1]$, so $2 - S_n$ is confined to $[1, 2)$, and bounding the denominator by its extremes leaves a geometric series that can be summed exactly.

Proposition 6.2 (Two-sided bounds). *Write $\varepsilon = 1 - 2p$ for the distance from criticality. Then for $p \in (0, \frac{1}{2})$,*

$$\frac{\varepsilon}{1 + \varepsilon} < A(p) < \frac{2\varepsilon}{1 + 2\varepsilon}, \quad (53)$$

that is,

$$\frac{1 - 2p}{2(1 - p)} < A(p) < \frac{2(1 - 2p)}{3 - 4p}. \quad (54)$$

The lower bound is exactly half the continuous-time constant of (47):

$$A(p) > \frac{1}{2}A_c(p). \quad (55)$$

Proof. Since $0 < S_n \leq 1$ for every n , we have $1 \leq 2 - S_n < 2$ and therefore $\frac{1}{2}r^n < r^n/(2 - S_n) \leq r^n$, with equality on the right only at $n = 0$. Summing over $n \geq 0$ and using $\sum_n r^n = 1/(1 - r) = 1/\varepsilon$,

$$1 + \frac{1}{2\varepsilon} < \frac{1}{A(p)} < 1 + \frac{1}{\varepsilon},$$

and inverting gives (53). Substituting $\varepsilon = 1 - 2p$ gives (54), whose lower bound is $(1 - 2p)/(2(1 - p)) = \frac{1}{2}A_c(p)$ by (47). $\square$

Two features of [proposition 6.2](#) deserve comment, because between them they account for the whole shape of the discrete curve.

At the far subcritical end, $p \rightarrow 0$, we have $\varepsilon \rightarrow 1$ and the lower bound tends to $\frac{1}{2}$. Since $A(p) \leq \frac{1}{2}$ by [proposition 6.3](#) below, the bound is asymptotically attained and $A(0^+) = \frac{1}{2}$. At the critical end, $p \rightarrow \frac{1}{2}^-$, we have $\varepsilon \rightarrow 0$, the lower bound behaves like ε and the upper like 2ε ; [section 6.3](#) shows that it is the upper bound that is attained there. So $A(p)$ runs from its lower bound at one end of the interval to its upper bound at the other, and no constant rescaling of A_c can reproduce it.

Proposition 6.3 (Parity bound). *For the binary Galton–Watson process started from a single individual, $A(p) \leq \frac{1}{2}$ for all $p \in [0, \frac{1}{2})$, with equality approached as $p \rightarrow 0$.*

Proof. Every individual leaves either 0 or 2 offspring, so $\mathbf{Z}_n$ is even for every $n \geq 1$. Conditional on survival we therefore have $\mathbf{Z}_n \geq 2$, and hence $\mathbb{E}(\mathbf{Z}_n \mid \mathbf{Z}_n > 0) \geq 2$ for every $n \geq 1$. Letting $n \rightarrow \infty$ and using (31) gives $1/A(p) \geq 2$. As $p \rightarrow 0$ the conditional population is 2 with probability tending to one, so the bound is attained in the limit. $\square$

Table 1: The constant $A(p)$ computed from the product (51) in 60-digit arithmetic, against the bounds of [proposition 6.2](#) and the near-critical asymptotic $2(1 - 2p)$ of [section 6.3](#). The bounds are loose in the middle of the range and tighten at both ends: at $p = 0.05$ the lower bound is already within 2.6% of A , whereas at $p = 0.4999$ it is the upper bound that is within 0.15%. The last two columns give the quasi-stationary mean and the number of factors of (51) needed for the product to reach full precision — the cost that [section 6.7](#) has to manage.

p	$\frac{1}{2}A_c(p)$	$A(p)$	$\frac{2(1 - 2p)}{3 - 4p}$	$2(1 - 2p)$	$1/A(p)$	iterations
0.05	0.473684	0.486180	0.642857	1.800000	2.0568	45
0.1	0.444444	0.469382	0.615385	1.600000	2.1305	64
0.2	0.375000	0.423895	0.545455	1.200000	2.3591	113
0.3	0.285714	0.353967	0.444444	0.800000	2.8251	201
0.4	0.166667	0.237647	0.285714	0.400000	4.2079	458
0.45	0.090909	0.145069	0.166667	0.200000	6.8933	966
0.48	0.038462	0.068011	0.074074	0.080000	14.7036	2 473
0.49	0.019608	0.036395	0.038462	0.040000	27.4764	4 965
0.499	0.001996	0.003944	0.003984	0.004000	253.519	48 992
0.4999	0.000200	0.000399	0.000400	0.000400	2504.65	478 905

6.3 Behaviour as $p \rightarrow \frac{1}{2}^-$

The series (52) also delivers the near-critical asymptotic, and it does so more transparently than the matching argument one is first tempted to try.

Fix $\varepsilon = 1 - 2p$ small. The summand $r^n/(2 - S_n)$ contains two decays on very different scales. The geometric factor $r^n = (1 - \varepsilon)^n$ decays on the scale $n \sim 1/\varepsilon$, which is large. The survival probability S_n , by contrast, stays close to its critical behaviour $S_n \sim 2/n$ across that whole range, and so has already fallen to near zero after $O(1)$ generations. For the overwhelming majority of terms carrying

weight in the sum we therefore have $S_n \approx 0$ and the summand is $\approx r^n/2$, whence

$$\sum_{n=0}^{\infty} \frac{r^n}{2 - S_n} \approx \frac{1}{2} \sum_{n=0}^{\infty} r^n = \frac{1}{2\varepsilon}.$$

So $1/A(p) \sim 1/(2\varepsilon)$, and

$$\boxed{A(p) \sim 2\varepsilon = 2(1 - 2p) \quad \text{as } p \rightarrow \frac{1}{2}^-} \quad (56)$$

Equivalently the quasi-stationary mean diverges like the reciprocal of the distance from criticality, $1/A(p) \sim 1/(2(1 - 2p))$. In particular $A(p)$ meets the axis at $p = \frac{1}{2}$ with gradient -4 , a fact used repeatedly below as a test of candidate formulae.

The same result can be reached through the continuum equation of [section 4.4](#), and the comparison is worth making because it shows where the difficulty in that route lies. With $\varepsilon = 1 - 2p$, the exact difference is

$$S_{n+1} - S_n = (2p - 1)S_n - pS_n^2 = -\varepsilon S_n - \frac{1}{2}S_n^2 + \frac{\varepsilon}{2}S_n^2,$$

and dropping the $O(\varepsilon S_n^2)$ term gives $dS/dn = -\varepsilon S - \frac{1}{2}S^2$. Setting $u = 1/S$ linearises this to $u' = \varepsilon u + \frac{1}{2}$, with solution

$$u(n) = Ce^{\varepsilon n} - \frac{1}{2\varepsilon}, \quad S(n) = \frac{1}{Ce^{\varepsilon n} - \frac{1}{2\varepsilon}}.$$

Matching $S_n \sim Ae^{-\varepsilon n}$ at large n and small ε identifies $A = 1/C$, and matching to the critical decay $S_n \sim 2/n$ in the appropriate double limit fixes $C \sim 1/(2\varepsilon)$, recovering (56). The series argument is preferable because it needs no second matching step, and the second matching step is the delicate one.

Remark 6.4 (Rate of approach). Numerically, $A(p)/2\varepsilon$ approaches 1 rather slowly. Evaluating (51) in high precision gives $A/2\varepsilon = 0.90987, 0.98612, 0.99814, 0.99977$ and 0.999972 at $\varepsilon = 2 \times 10^{-2}$ down to 2×10^{-6} . Fitting the deficit gives $1 - A/2\varepsilon \approx \varepsilon(\ln(1/\varepsilon) + c)$ with $c \approx 0.78$, so

$$A(p) = 2\varepsilon \left[1 + O(\varepsilon \ln(1/\varepsilon)) \right].$$

The logarithmic correction is why (56) is a good practical surrogate only very close to criticality, and it is what fixes the crossover point chosen in [section 6.7](#).

6.4 Why the discrete and continuous constants differ

We can now settle the question raised by [figure 9](#). The two constants are

$$A_c(p) = \frac{1 - 2p}{1 - p} \quad (\text{continuous time}), \quad A(p) = \lim_{n \rightarrow \infty} \frac{S_n}{(2p)^n} \quad (\text{discrete time}),$$

and the observation was that the first starts at 1 where the second starts at $\frac{1}{2}$, yet halving the first does not superpose the curves.

[Proposition 6.2](#) explains this completely. What holds is not $A = \frac{1}{2}A_c$ but the strict inequality $A > \frac{1}{2}A_c$, and equality is approached only as $p \rightarrow 0$. Combining (47) and (56), the ratio runs monotonically across the interval,

$$\frac{A(p)}{A_c(p)} \longrightarrow \frac{1}{2} \quad (p \rightarrow 0), \quad \frac{A(p)}{A_c(p)} \longrightarrow 1 \quad (p \rightarrow \frac{1}{2}^-), \quad (57)$$

taking the values 0.503, 0.528, 0.619, 0.798, 0.928 and 0.988 at $p = 0.01, 0.1, 0.3, 0.45, 0.49, 0.499$. No constant factor will do, because the ratio is not constant.

The two endpoints have clean interpretations, and they are different in kind.

At $p \rightarrow 0$ the factor of two is a counting effect. In the binary discrete model an individual leaves either nothing or two offspring, so the surviving population is always even ([proposition 6.3](#)); deep in the subcritical regime, conditioning on survival means conditioning on a population of exactly two, and the quasi-stationary mean is 2. In the continuous-time model a survivor is a single individual that simply has not yet died, so the quasi-stationary mean is 1. The reciprocals are $\frac{1}{2}$ and 1, and there is the discrepancy.

At $p \rightarrow \frac{1}{2}^-$ the factor disappears, because both constants are governed by the same critical scaling. Expanding (47) near $p = \frac{1}{2}$ gives $A_c(p) = (1 - 2p)/(1 - p) \rightarrow 2(1 - 2p)$, which is precisely (56). Near criticality the discreteness of the generational clock stops mattering; what survives is the universal 2ε behaviour, and the two models agree to leading order.

The discrete curve is therefore squeezed between $\frac{1}{2}A_c$ and its own critical asymptote, hugging the former at one end of the interval and the latter at the other. That is the whole content of [figure 9](#).

6.5 Searching for a closed form

Before the connection of [section 6.6](#) was appreciated, a number of methods were tried in the hope of an elementary formula for $A(p)$. Two are worth recording, partly for the negative result and partly because the way they failed was itself informative.

The first was the continuum route already described in [section 4.4](#): treat the map as an ODE in the large-generation limit and solve it. That gives the shape of the decay, but it leaves the constant of integration undetermined, which is to say it leaves $A(p)$ undetermined.

The second was symbolic regression by genetic algorithm. The concept is simple: generate fifty candidate functions at random, measure how closely each reproduces the target curve, retain the best five, and build a fresh population of fifty by varying those five; then measure again, and repeat the cycle of variation and selection many times over [\[6, 7\]](#).

The first implementation searched a space of polynomials, partly on the hunch that the answer might take that form and partly because such functions are simple to encode. In the minimal case a genome is a coefficient list $[a_0, \dots, a_k]$ for

$$a_0 + a_1p + a_2p^2 + \dots + a_kp^k$$

up to some chosen maximum order k , with the initial population drawn uniformly at random. It quickly became apparent that many substantially different formulae produce curves visually indistinguishable from the target — but we wanted the exact one. On the supposition that a true formula might have rational coefficients, a second version searched

$$\frac{a_0}{b_0} + \frac{a_1}{b_1}p + \dots + \frac{a_k}{b_k}p^k, \quad [a_i], [b_i] \in \mathbb{Z}^{k+1},$$

and a third extended the space to ratios of two such polynomials, with genome in $\mathbb{Z}^{4(k+1)}$. After numerous searches the best that emerged merely fitted the high-precision numerical data well:

$$\hat{A}_1(p) = \frac{\frac{1}{2}(1 - 2p)(1 + 2p)}{1 + \frac{1}{2}p - \frac{5}{2}p^2 - \frac{6}{5}p^3 + \frac{6}{5}p^4}. \quad (58)$$

This passes through $(0, \frac{1}{2})$ and $(\frac{1}{2}, 0)$ and looks convincing to the eye, but it fails the diagnostic of (56): differentiating at $p = \frac{1}{2}$ gives a gradient of $-2/0.55 = -3.636$ rather than the required -4 .

Two years after this work was set aside it was revisited with more capable tooling, using a symbolic tree search over progressively nested elementary functions of all kinds. The conclusion was the same. Many functions of widely varying form resemble the target closely, and no satisfactory simple formula emerges even when solutions with fewer components are explicitly favoured. Two arbitrary picks from the sea of similarly performing candidates are

$$\begin{aligned} \hat{A}_2(p) = & 0.31696448624690005 \ln \left(\left| \cos \left(\left(\frac{1}{3.19801133824532} \right)^{1/4} \right. \right. \right. \\ & \left. \left. \left. + p \left[\sin \left(\cos(|p| (0.621865696665606 - p) \cos(e^{e^p})) \right) + |p| \right] \right) \right| \right) + 0.5982282661889831 \end{aligned} \quad (59)$$

and, more concisely,

$$\hat{A}_3(p) = \frac{-0.616647881739505 e^{-0.552605335919693 e^p}}{\sin(\cos p) - |p|} + 0.921964323679771. \quad (60)$$

(59) is quoted after removing redundancies such as a factor p/p that the program had introduced — neutral mutations, in effect, carrying no fitness cost and so never selected against.

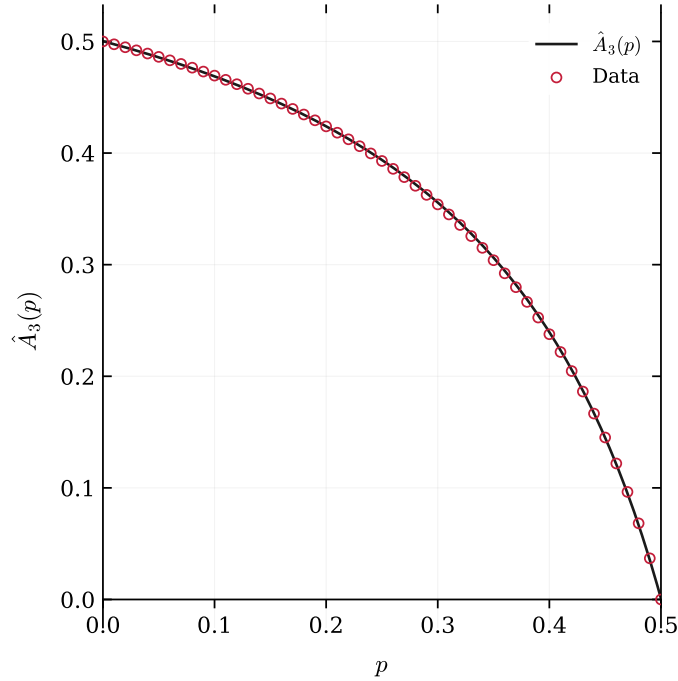


Figure 11: The symbolic-regression candidate $\hat{A}_3(p)$ of (60) (solid) against numerically computed values of $A(p)$ (red circles). The agreement is excellent to the eye across the whole interval, which is exactly the problem: the fit is not evidence of correctness. $\hat{A}_3$ misses the critical point, giving $\hat{A}_3(\frac{1}{2}) = 9.1 \times 10^{-4}$ rather than 0, and arrives there with gradient -3.63 instead of -4 . Errors of this size are invisible here but are fatal for $1/A$ near criticality.

The recurring defect is the same in every case. The candidates either fail to pass through $(\frac{1}{2}, 0)$ or fail to arrive there with gradient -4 , and near criticality that matters enormously: as $A \rightarrow 0$, an error in A of vanishing absolute size produces an $O(1)$ — eventually unbounded — error in the

quasi-stationary mean $1/A$. A formula can look perfect on a plot of A and be useless on a plot of $1/A$.

A pragmatic patch is to splice the asymptotic onto a fitted curve, as in

$$\hat{A}_4(p) = \begin{cases} -0.04011483983619545 e^{2p} + 0.6079994336528085, & p < 0.499962, \\ 2 - 4p, & p \geq 0.499962, \end{cases} \quad (61)$$

which does work. But if one is going to splice, it is better to splice onto something exact, and [section 6.7](#) does so.

The deeper lesson from these searches points directly at the next subsection. That an enormous number of very different formulae can behave almost identically over a bounded interval is not merely an inconvenience of the method; it is a symptom. Curve-fitting cannot distinguish a function that has a closed form from one that does not.

6.6 The Koenigs connection

It was only some time after the search had been abandoned that a dynamical argument emerged explaining why no elementary closed form should be expected. In short, $A(p)$ is a value of the Koenigs linearising function of the logistic map, and that function is known to be non-elementary throughout the parameter range we need.

Definition 6.5 (Koenigs function). Let $f(z) = rz(1 - z)$ have a fixed point at $z = 0$ with multiplier $r = f'(0)$, and suppose $|r| \notin \{0, 1\}$. Koenigs [8] proved that there exists a unique function ψ_r , analytic near the origin, satisfying the Schröder equation

$$\psi_r(f(z)) = r \psi_r(z), \quad \psi_r(0) = 0, \quad \psi'_r(0) = 1. \quad (62)$$

The Koenigs function is a change of coordinate that straightens the dynamics: in the ψ coordinate the nonlinear map f becomes multiplication by r ([figure 12](#)). Here $f(z) = rz(1 - z)$ is exactly (49), with $z = w_n$ and $f(z) = w_{n+1}$, and the multiplier is $r = 2p \in (0, 1)$ — an attracting fixed point, which is the classical Koenigs setting.

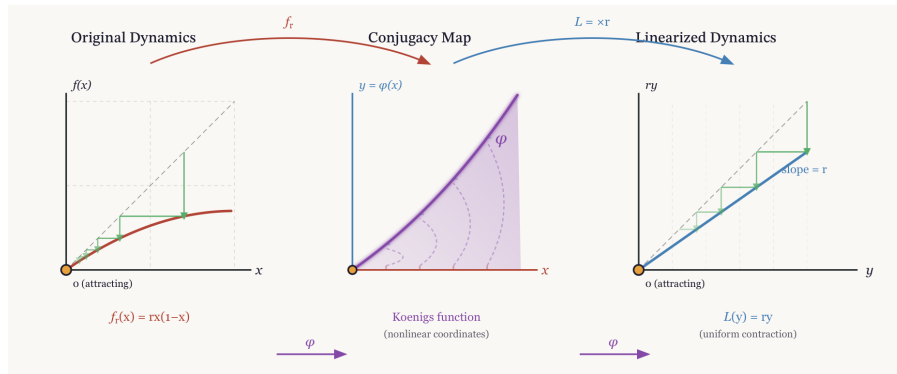


Figure 12: Koenigs linearisation of the logistic map near an attracting fixed point. The nonlinear dynamics $x \mapsto f_r(x) = rx(1 - x)$ are conjugated by ψ_r to the linear map $y \mapsto ry$. The diagram displays the functional relation $\psi_r(f_r(x)) = r\psi_r(x)$ and the role of ψ_r as the coordinate change that straightens the dynamics.

Proposition 6.6. For $p \in (0, \frac{1}{2})$ and $r = 2p$,

$$A(p) = 2\psi_r\left(\frac{1}{2}\right). \quad (63)$$

Proof. Iterating (62) along the orbit $w_0, w_1, w_2, \dots$ of (49) gives $\psi(w_1) = r\psi(w_0)$, hence $\psi(w_0) = \psi(w_1)/r$; similarly $\psi(w_1) = \psi(w_2)/r$, so $\psi(w_0) = \psi(w_2)/r^2$; and by induction

$$\psi(w_0) = \frac{\psi(w_n)}{r^n} \quad \text{for every } n. \quad (64)$$

Since (64) holds for all n we may pass to the limit, $\psi(w_0) = \lim_{n \rightarrow \infty} \psi(w_n)/r^n$. The normalisation $\psi(0) = 0$, $\psi'(0) = 1$ means $\psi(z)/z \rightarrow 1$ as $z \rightarrow 0$, and $w_n = S_n/2 \rightarrow 0$ because $r < 1$; replacing $\psi(w_n)$ by w_n in the limit is therefore legitimate, and

$$\psi(w_0) = \lim_{n \rightarrow \infty} \frac{w_n}{r^n}.$$

Comparing with (50) and using $w_0 = S_0/2 = \frac{1}{2}$ gives $A(p) = 2\psi_r(\frac{1}{2})$. $\square$

When is ψ_r elementary?

The answer is: essentially never. The logistic family is affinely conjugate to the canonical quadratic family $P_c(z) = z^2 + c$; substituting $x = -z/r + \frac{1}{2}$ in f_r gives P_c with

$$c = \frac{2r - r^2}{4}, \quad (65)$$

under which the interval $r \in (0, 1)$ maps to $c \in (0, \frac{1}{4})$ (figure 13). The relevance of this is a rigidity theorem of Becker and Bergweiler [9]: for a rational map R of degree at least two with a fixed point of multiplier λ , $0 < |\lambda| < 1$, the associated Koenigs linearisation satisfies an algebraic differential equation only when R is analytically conjugate to a monomial map $z \mapsto z^{\pm d}$ or to a Chebyshev polynomial $\pm T_d$. For the quadratic family these exceptional cases are exactly $c = 0$ and $c = -2$, corresponding under (65) to $r = 2$ and $r = 4$.

Every elementary function in the sense of Liouville — algebraic functions, exponentials, logarithms and their finite combinations — satisfies an algebraic differential equation, so a function that satisfies none, a *hypertranscendental* function, cannot be elementary. And

$$\left(0, \frac{1}{4}\right) \cap \{0, -2\} = \emptyset,$$

so no r in our range is exceptional: for each fixed $r \in (0, 1)$ the Koenigs function ψ_r is hypertranscendental, and in particular not elementary.

The two exceptional parameters are worth writing out, both as a check on the machinery and because they show how special the solvable cases are.

The monomial case, $r = 2$. Here $f_2(x) = 2x(1 - x)$, and the substitution $u = 1 - 2x$ gives

$$u_{n+1} = 1 - 2(2x_n(1 - x_n)) = 1 - 4x_n + 4x_n^2 = (1 - 2x_n)^2 = u_n^2.$$

The map $u \mapsto u^2$ is linearised by the logarithm, since $\ln(u^2) = 2 \ln u$. Returning to x and imposing the normalisation of definition 6.5,

$$\psi_2(x) = -\frac{1}{2} \ln(1 - 2x), \quad (66)$$

which is readily verified: $\psi_2(f_2(x)) = -\frac{1}{2} \ln((1 - 2x)^2) = 2[-\frac{1}{2} \ln(1 - 2x)] = 2\psi_2(x)$.

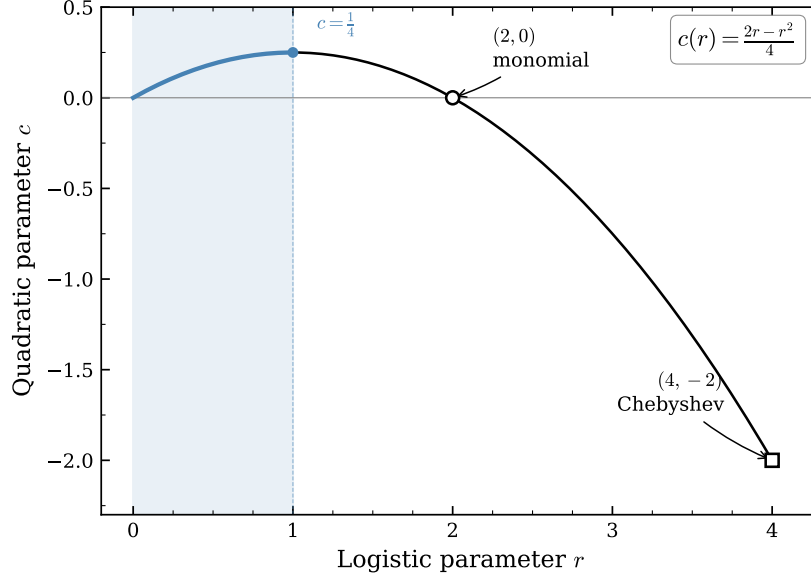


Figure 13: The affine parameter correspondence $c(r) = (2r - r^2)/4$ between the logistic family $f_r(x) = rx(1 - x)$ and the quadratic family $P_c(z) = z^2 + c$. The shaded segment is $r \in (0, 1)$, the subcritical branching window, which maps to $c \in (0, \frac{1}{4})$. The two integrable parameters, $r = 2$ ($c = 0$, monomial) and $r = 4$ ($c = -2$, Chebyshev), lie outside it.

The Chebyshev case, $r = 4$. Here $f_4(x) = 4x(1 - x)$ is conjugate to $T_2(z) = 2z^2 - 1$ by a trigonometric substitution. Putting $x = \sin^2 \theta$,

$$f_4(x) = 4 \sin^2 \theta \cos^2 \theta = (2 \sin \theta \cos \theta)^2 = \sin^2(2\theta),$$

so the map is a doubling $\theta \mapsto 2\theta$ in the θ coordinate. The multiplier at the origin is 4 rather than 2, which suggests that the linearisation involves θ^2 ; and indeed

$$\psi_4(x) = (\arcsin \sqrt{x})^2. \quad (67)$$

Using $\arcsin(2u\sqrt{1-u^2}) = 2 \arcsin u$ with $u = \sqrt{x}$ gives $\psi_4(f_4(x)) = (2 \arcsin \sqrt{x})^2 = 4\psi_4(x)$; and as $x \rightarrow 0$, $\arcsin \sqrt{x} \approx \sqrt{x}$, so $\psi_4(x) \approx x$ and $\psi_4'(0) = 1$.

Remark 6.7 (Attracting versus repelling). At $r = 2$ and $r = 4$ the multiplier at the origin satisfies $|r| > 1$, so the origin is *repelling*. The functions (66) and (67) are therefore Poincaré-type linearisations rather than attracting-Koenigs maps, and neither parameter lies in the branching window $r = 2p \in [0, 1)$ in any case. They are recorded to show what a solvable case looks like, not because they bear on $A(p)$.

What this does and does not establish

It is worth being precise about the status of the conclusion, because it is easy to overstate.

What is established is that for each *fixed* $r \in (0, 1)$ the function $z \mapsto \psi_r(z)$ is hypertranscendental. Combined with [proposition 6.6](#), this says that $A(p)$ is a value of a non-elementary function, evaluated at the fixed argument $z = \frac{1}{2}$, for every p in the subcritical range. It is a strong indication that the search of [section 6.5](#) was doomed, and it explains the phenomenology of that search: the object being fitted is defined only by the dynamics that generate it.

What is *not* established is that the map $p \mapsto A(p)$ admits no elementary description. Hypertranscendence of ψ_r in its argument z is a statement about one variable, whereas $A(p) = 2\psi_{2p}(\frac{1}{2})$ varies the parameter and freezes the argument, and the two questions are logically distinct. A rigorous negative result for $p \mapsto A(p)$ would require a transcendence statement about the values rather than the functions, and none is claimed here. The honest summary is that no closed form is known, that the structural reason for expecting none is the one just given, and that the numerical evidence of [section 6.5](#) is entirely consistent with there being none.

It helps to see where in the quadratic family our parameters sit, because the picture makes the difficulty look structural rather than accidental. The main cardioid of the Mandelbrot set — the set of c for which $z^2 + c$ has an attracting fixed point — is parametrised by that point's multiplier λ through $c = \lambda/2 - \lambda^2/4$, which is exactly (65) with $\lambda = r$. The branching window $r \in (0, 1)$ is therefore the real diameter of the main cardioid, traversed from the centre $c = 0$ out to the parabolic cusp $c = \frac{1}{4}$, where $\lambda = 1$ and the process turns critical. [Figure 14](#) places the segment in context; the two integrable parameters $c = 0$ and $c = -2$ are marked, and the open window contains neither.

Throughout the window the Julia set is a quasicircle bounding the basin of the attracting fixed point,¹ and it is a genuine fractal for every c in the interval. The Koenigs function is the conformal coordinate that linearises the dynamics inside that basin, so its natural domain is bounded by exactly this curve. A function whose domain of definition has no smooth description is not a promising candidate for a closed form, and $A(p)$, being one of its values, inherits the difficulty. The contrast with the integrable cases is sharp: at $c = 0$ the Julia set degenerates to the unit circle, and it is precisely there that the linearisation becomes elementary, (66).

6.7 Computing $A(p)$ in practice

In the absence of an exact formula we are left with approximation, and the two regimes of [section 6.2](#) indicate how to split the work.

Over most of the interval, direct iteration is entirely satisfactory. The infinite product (51) converges quickly, its partial products bound $A(p)$ from above at every order, and modest numbers of iterations give many digits. The difficulty is confined to a neighbourhood of $p = \frac{1}{2}$, and it has two parts. First, as $A(p) \rightarrow 0$ the quantity of interest $1/A$ diverges, so a fixed absolute error in A becomes an unbounded error in the quasi-stationary mean. Second — and this is the binding constraint — as the process is defined ever closer to criticality its lifetime, though still finite, grows enormously, and S_n recedes to zero ever more slowly. The iteration must be carried to a generation at which S_n is essentially unchanging, and [table 1](#) shows what that costs: about 10^2 iterations at $p = 0.2$, 10^3 at $p = 0.45$, but nearly 5×10^5 at $p = 0.4999$.

The remedy is to hand the near-critical region to the asymptotic (56), which is exact in the limit and cheap everywhere:

$$A(p) \approx \begin{cases} \frac{1}{2} \prod_{k=1}^N \left(1 - \frac{S_k}{2}\right), & p < p_\star, \quad N \gg 1, \\ 2 - 4p, & p \geq p_\star, \end{cases} \quad (68)$$

with the crossover $p_\star$ placed just below $\frac{1}{2}$. [Remark 6.4](#) says where to put it: the relative error of the asymptotic is about $\varepsilon \ln(1/\varepsilon)$, so at $p_\star = 0.4999$ ($\varepsilon = 2 \times 10^{-4}$) the asymptotic is already accurate

¹An interactive visualiser accompanying this chapter plots ψ_r over the complex plane by the product form $\psi_r(z) = z \prod_{n \geq 0} (1 - f_r^{(n)}(z))$ — the same telescoping identity used in [section 6.1](#) — colouring by the phase of ψ_r so that the linearised coordinate grid and the fractal basin boundary are both visible ([Koenigs_plotter/gem_p1.html](#)).

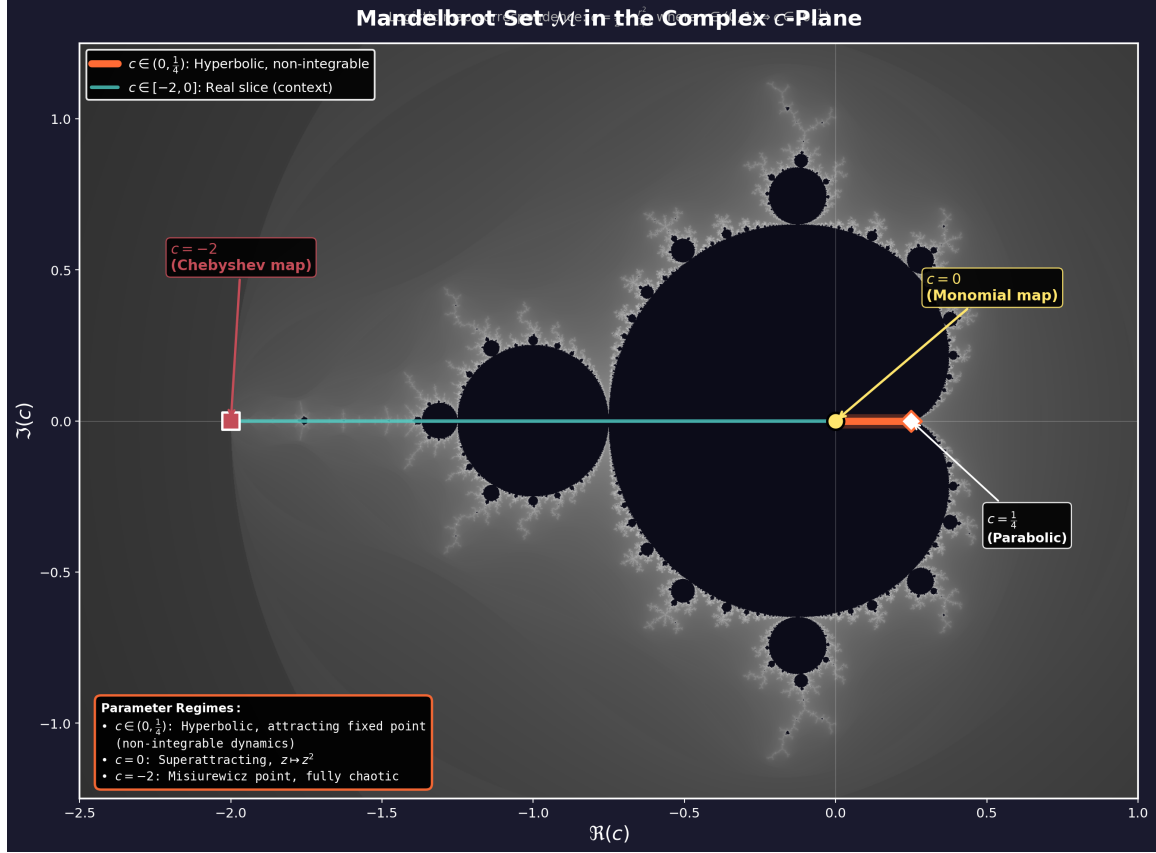


Figure 14: The Mandelbrot set in the complex c -plane, with the interval $c \in (0, \frac{1}{4})$ highlighted. Under the correspondence $c = (2r - r^2)/4$ of (65) this is the image of the subcritical branching window $r = 2p \in (0, 1)$, and it forms the real diameter of the main cardioid, running from the centre $c = 0$ to the parabolic cusp $c = \frac{1}{4}$, which is criticality. The two integrable parameters are marked: $c = 0$ (monomial, reached in logistic coordinates at $r = 2$) and $c = -2$ (Chebyshev, $r = 4$), the latter at the far tip of the real slice. Note that $c(r) = c(2 - r)$, so each $c \in (0, \frac{1}{4})$ is the image of two logistic parameters, r and $2 - r$; these correspond to the two fixed points of $z^2 + c$, with multipliers r and $2 - r$, and it is the attracting one, $r < 1$, that governs $A(p)$.

to about two parts in 10^3 , while the product still converges in a few hundred thousand iterations. Any p_* in the range 0.499–0.4999 gives a scheme accurate to better than one per cent throughout, and the two branches can be blended over a narrow window if continuity of the derivative matters.

This is the recipe used for the numerical values quoted throughout the chapter, including those in figure 5 and table 1.

7 Absorption models and the method of characteristics

The chapter now changes tack. Everything so far has concerned a single population and the chance that it survives; the remainder develops the analytical machinery for a second kind of question, in which particles move between compartments and one wants the joint distribution of the counts on either side.

The motivation is a modelling choice that arises as soon as more than one compartment is present.

Earlier work describes the birth–death dynamics of particles held inside a single compartment that can rupture. Extending that description to several compartments sharing a medium requires a decision about what happens when the first of them ruptures, because the shared medium then acquires a potentially large cohort of particles available for subsequent uptake by the remaining compartments. The analysis so far has assumed *immediate transfer*: the entire contents of a rupturing compartment relocate at once into another compartment. That is a strong assumption. A more realistic alternative is an *absorption period*, in which absorption events redistribute the released cohort among the remaining compartments, and birth, death and rupture resume only when every particle has been taken up. Working towards such models, we treat here the case in which a cohort of particles enters a single compartment.

Three models are treated in increasing order of difficulty. In the first two, particles are conserved or only lost, the state space is finite, and an independence argument delivers the generating function without solving a partial differential equation at all. In the third, replication inside the compartment opens an infinite state space and the independence shortcut fails, so the PDE must be confronted directly. We therefore develop the method of characteristics on the second model, where the answer is already known, and then apply it to the third.

7.1 Absorption only

The simplest absorption process has a single compartment that takes in particles, with no subsequent dynamics — no birth and no death of particles inside it.

Write $\mathbf{X}_t$ and $\mathbf{Y}_t$ for the number of particles inside and outside the compartment respectively; the same labelling is used for every version of the single-compartment model below. Events are stochastic with exponentially distributed inter-event times, which follows from the assumption that event rates do not depend on time, so the process $\{(\mathbf{X}_t, \mathbf{Y}_t) : t \in [0, \infty)\}$ is a CTMC in the sense of [section 2.2](#). With only absorption to consider, the model is specified by the following outcome likelihoods over an infinitesimal duration Δt :

$$\begin{aligned} \Pr\{\Delta \mathbf{X} = 1, \Delta \mathbf{Y} = -1 \mid \mathbf{Y}_t = j\} &= j\alpha\Delta t, & (\text{absorption}) \\ \Pr\{\Delta \mathbf{X} = 0, \Delta \mathbf{Y} = 0 \mid \mathbf{Y}_t = j\} &= 1 - j\alpha\Delta t, & (\text{no event}) \end{aligned}$$

where $\Delta \mathbf{X} = \mathbf{X}_{t+\Delta t} - \mathbf{X}_t$ and similarly for $\mathbf{Y}$. The factor j matters: each exterior particle is independently subject to the same probability $\alpha\Delta t$ of being taken up, so the population rate is linear in the exterior count, exactly as in [\(14\)](#). The initial condition is $\mathbf{X}_0 = 0, \mathbf{Y}_0 = N$ — a starting population of census N occupying the exterior medium.

Writing $p_{i,j}(t) = \Pr\{\mathbf{X}_t = i, \mathbf{Y}_t = j\}$ for the state probabilities, the forward equations [\(8\)](#) read

$$\frac{dp_{i,j}}{dt} = \alpha(j+1)p_{i-1,j+1} - \alpha j p_{i,j}. \quad (69)$$

The route from here to a generating function is the one set out in [section 2.3](#), and because it is used three times below it is worth doing once in full. Multiply [\(69\)](#) by $x^i y^j$ and sum over all states with $p_{i,j}(t) \neq 0$:

$$\sum_{i=0}^N \sum_{j=0}^N \frac{\partial p_{i,j} x^i y^j}{\partial t} = \alpha \sum_{i=1}^{N+1} \sum_{j=-1}^{N-1} (j+1) p_{i-1,j+1} x^i y^j - \alpha \sum_{i=0}^N \sum_{j=0}^N j p_{i,j} x^i y^j. \quad (70)$$

The sums are finite because $p_{i,j}(t) = 0$ for all t whenever $i + j \neq N$: particles in this model are neither created nor destroyed. Shifting indices in the first double sum, so that $p_{i-1,j+1}$ becomes $p_{i,j}$

at the cost of one power of x and one of y , gives

$$\sum_{i=0}^N \sum_{j=0}^N \frac{\partial p_{i,j} x^i y^j}{\partial t} = \alpha \left(x \sum_{i=0}^N \sum_{j=0}^N j p_{i,j} x^i y^{j-1} - y \sum_{i=0}^N \sum_{j=0}^N j p_{i,j} x^i y^{j-1} \right),$$

and both remaining sums are the partial derivative of (13) with respect to y . Recognising $G(x, y, t)$ therefore turns the whole system into one equation,

$$\frac{\partial G(x, y, t)}{\partial t} = \alpha(x - y) \frac{\partial G(x, y, t)}{\partial y}, \quad G(x, y, 0) = y^N. \quad (71)$$

This first-order linear initial value problem proves surprisingly awkward. The method of characteristics applies in principle to equations in three variables, but applying it here led at first only to mystification. As it happens the solution can be obtained by other means, and it is worth obtaining it that way both because the answer is needed and because it provides a check on the characteristics machinery developed in [section 7.3](#).

Exploiting independence

The function G above is specific to $\mathbf{X}_0 = 0$, $\mathbf{Y}_0 = N$. Define more generally

$$G_n(x, y, t) = \sum_i \sum_j \Pr \{ \mathbf{X}_t = i, \mathbf{Y}_t = j \mid \mathbf{X}_0 = 0, \mathbf{Y}_0 = n \} x^i y^j.$$

In a process of pure absorption the n particles are taken up one at a time, and a single particle's behaviour does not depend on the movements of its neighbours. Both counts therefore decompose into sums of n single-particle variables,

$$\mathbf{X}_t = \mathbf{X}_t^{(1)} + \cdots + \mathbf{X}_t^{(n)}, \quad \mathbf{Y}_t = \mathbf{Y}_t^{(1)} + \cdots + \mathbf{Y}_t^{(n)},$$

and (11) applies in its bivariate form:

$$G_n(x, y, t) = \mathbb{E} \left(\prod_{k=1}^n x^{\mathbf{X}_t^{(k)}} y^{\mathbf{Y}_t^{(k)}} \right) = \prod_{k=1}^n \mathbb{E} \left(x^{\mathbf{X}_t^{(k)}} y^{\mathbf{Y}_t^{(k)}} \right) = (G_1(x, y, t))^n. \quad (72)$$

So it is enough to solve the one-particle problem. With $\mathbf{X}_0 = 0$, $\mathbf{Y}_0 = 1$ the particle is eventually absorbed and that is the final state of the system: no other states are reachable, so $p_{i,j}(t) = 0$ for all t whenever $i + j \neq 1$. The forward equations reduce to two,

$$\frac{dp_{0,1}(t)}{dt} = -\alpha p_{0,1}(t), \quad \frac{dp_{1,0}(t)}{dt} = \alpha p_{0,1}(t),$$

giving $p_{0,1}(t) = e^{-\alpha t}$ and $p_{1,0}(t) = 1 - e^{-\alpha t}$ ([figure 15](#)).

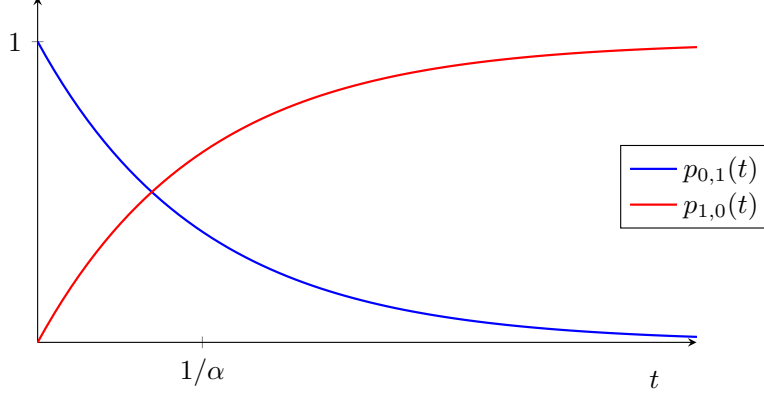


Figure 15: State probabilities for the single-particle absorption-only model. The particle begins outside the compartment ($p_{0,1}$, decaying exponentially at rate α) and ends inside it ($p_{1,0}$). There is nowhere else for it to go, so the two curves sum to one at every time.

These being the only two non-zero state probabilities of the single-particle system,

$$G_1(x, y, t) = p_{1,0}(t)x + p_{0,1}(t)y = (1 - e^{-\alpha t})x + e^{-\alpha t}y,$$

and (72) gives the generating function for any initial census,

$$G_n(x, y, t) = \left((1 - e^{-\alpha t})x + e^{-\alpha t}y \right)^n. \quad (73)$$

This satisfies (71) together with the initial condition $G_n(x, y, 0) = y^n$, and its form is recognisable: $\mathbf{X}_t \sim \text{Binomial}(n, 1 - e^{-\alpha t})$, as it must be, since the n particles are absorbed independently and each has been taken up by time t with probability $1 - e^{-\alpha t}$.

7.2 Absorption–death

A particle resident in a compartment may die. Keeping $\mathbf{X}_t$ and $\mathbf{Y}_t$ for the interior and exterior counts, death enters as a third channel:

$$\begin{aligned} \Pr \{ \Delta \mathbf{X} = 1, \Delta \mathbf{Y} = -1 \mid \mathbf{Y}_t = j \} &= j\alpha \Delta t, & (\text{absorption}) \\ \Pr \{ \Delta \mathbf{X} = -1, \Delta \mathbf{Y} = 0 \mid \mathbf{X}_t = i \} &= i\mu \Delta t, & (\text{death}) \\ \Pr \{ \Delta \mathbf{X} = 0, \Delta \mathbf{Y} = 0 \mid \mathbf{Y}_t = j, \mathbf{X}_t = i \} &= 1 - (j\alpha + i\mu) \Delta t, & (\text{no event}) \end{aligned}$$

with forward equations

$$\frac{dp_{i,j}}{dt} = \alpha(j+1)p_{i-1,j+1} + \mu(i+1)p_{i+1,j} - (\alpha j + \mu i)p_{i,j}. \quad (74)$$

The derivation of section 7.1 runs unchanged on the extra terms and produces

$$\frac{\partial G(x, y, t)}{\partial t} = \mu(1-x) \frac{\partial G(x, y, t)}{\partial x} + \alpha(x-y) \frac{\partial G(x, y, t)}{\partial y}, \quad G(x, y, 0) = y^N, \quad (75)$$

and the same difficulty. Once again, though, individual particles remain independent in their movements, so (72) still applies.

Without birth events the number of particles cannot exceed its initial value, so for an isolated particle the generating function has just three terms,

$$G_1(x, y, t) = p_{0,0}(t) + p_{1,0}(t)x + p_{0,1}(t)y,$$

corresponding to a particle that is dead, absorbed, or still outside. The state probabilities follow from

$$\frac{dp_{0,1}(t)}{dt} = -\alpha p_{0,1}(t), \quad \frac{dp_{1,0}(t)}{dt} = \alpha p_{0,1}(t) - \mu p_{1,0}(t),$$

whence

$$p_{0,1}(t) = e^{-\alpha t}, \quad p_{1,0}(t) = \frac{\alpha}{\mu - \alpha} (e^{-\alpha t} - e^{-\mu t}), \quad p_{0,0}(t) = 1 - p_{0,1}(t) - p_{1,0}(t), \quad (76)$$

plotted in [figure 16](#).

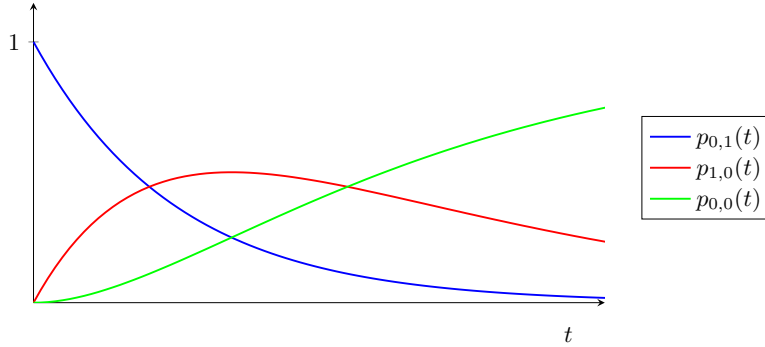


Figure 16: State probabilities for the single-particle absorption–death model, (76). The particle starts outside ($p_{0,1}$, blue), is absorbed into the compartment ($p_{1,0}$, red, which rises and then decays as death takes over), and finally dies ($p_{0,0}$, green, absorbing).

Applying (72) gives the general generating function

$$G_n(x, y, t) = \left(1 - \frac{\mu}{\mu - \alpha} e^{-\alpha t} + \frac{\alpha}{\mu - \alpha} e^{-\mu t} + \frac{\alpha}{\mu - \alpha} (e^{-\alpha t} - e^{-\mu t})x + e^{-\alpha t}y\right)^n. \quad (77)$$

7.3 The method of characteristics, tested on absorption–death

Before continuing to the case in which replication occurs inside the compartment and the independence shortcut is unavailable, it is worth revisiting absorption–death from the point of view of its PDE. Having (77) already in hand makes it an ideal test case.²

Introduce a change of variables $(x, y, t) \rightarrow (s, \rho, \tau)$, under which

$$G(x, y, t) = G(x(s, \rho, \tau), y(s, \rho, \tau), t(s, \rho, \tau)) \equiv G(s, \rho, \tau).$$

The method works by seeking solutions along the characteristic curves of constant s and ρ , on which G is a function of τ alone. To fix the parametrisation we impose an initial surface,

$$x(s, \rho, 0) = s, \quad y(s, \rho, 0) = \rho, \quad t(s, \rho, 0) = 0. \quad (78)$$

²I am grateful to Benjamin Sharp for the advice on applying the method of characteristics to three-variate problems that unlocked this section.

With G a function of τ only along a characteristic, the chain rule gives

$$\frac{dG}{d\tau} = \frac{\partial G}{\partial t} \frac{dt}{d\tau} + \frac{\partial G}{\partial x} \frac{dx}{d\tau} + \frac{\partial G}{\partial y} \frac{dy}{d\tau},$$

which is to be compared with (75) written with everything on one side,

$$0 = -\frac{\partial G}{\partial t} + \mu(1-x)\frac{\partial G}{\partial x} + \alpha(x-y)\frac{\partial G}{\partial y}.$$

Comparing coefficients yields the characteristic system

$$\frac{dG}{d\tau} = 0, \tag{79}$$

$$\frac{dt}{d\tau} = -1, \tag{80}$$

$$\frac{dx}{d\tau} = \mu(1-x), \tag{81}$$

$$\frac{dy}{d\tau} = \alpha(x-y). \tag{82}$$

Four ordinary differential equations have replaced one partial differential equation, and they are solved in the order written, each supplying what the next needs.

The equation for G , (79). G is constant along characteristics, $G = \varphi(s, \rho)$. Under (78) the initial condition $G(x, y, 0) = y^N$ becomes $G(s, \rho, 0) = \rho^N$, so $\varphi(s, \rho) = \rho^N$ and

$$G(s, \rho, \tau) = \rho^N. \tag{83}$$

The whole problem thus reduces to expressing ρ in terms of (x, y, t) .

The equation for t , (80). Integrating gives $t = -\tau + c(s, \rho)$, and (78) fixes $c = 0$:

$$t = -\tau. \tag{84}$$

The characteristic parameter runs backwards in time, which is what allows the initial surface to be reached from any (x, y, t) .

The equation for x , (81). Separating variables gives $1-x = c(s, \rho)e^{-\mu\tau}$, and (78) gives $c = 1-s$:

$$1-x = (1-s)e^{-\mu\tau}. \tag{85}$$

The equation for y , (82). Substituting (85) makes the y equation linear,

$$\frac{dy}{d\tau} = \alpha(1 - (1-s)e^{-\mu\tau} - y),$$

and solving with integrating factor $e^{\alpha\tau}$,

$$y = 1 - \frac{\alpha}{\alpha - \mu}(1-s)e^{-\mu\tau} + c(s, \rho)e^{-\alpha\tau}.$$

Applying (78) gives $c(s, \rho) = \rho + \frac{\alpha}{\alpha - \mu}(1-s) - 1$, so

$$y = 1 - \frac{\alpha}{\alpha - \mu}(1-s)e^{-\mu\tau} + \left(\rho + \frac{\alpha}{\alpha - \mu}(1-s) - 1\right)e^{-\alpha\tau}. \tag{86}$$

Recovering the solution. Rearranging (86) for ρ , then substituting s from (85) and $\tau = -t$ from (84), gives

$$\rho = 1 - \frac{\mu}{\mu - \alpha} e^{-\alpha t} + \frac{\alpha}{\mu - \alpha} e^{-\mu t} + \frac{\alpha}{\mu - \alpha} (e^{-\alpha t} - e^{-\mu t}) x + e^{-\alpha t} y,$$

which is exactly the bracket of (77); substituting into (83) reproduces that solution. The two routes agree, and the machinery is ready for the case where only one of them is available.

7.4 Absorption–birth–death

Introducing replication for particles inside the compartment, we shall see that although the independence property is still present it no longer serves as a shortcut. Given a single initial particle, birth events make infinitely many states reachable, so $G_1(x, y, t)$ is an infinite sum and there is nothing to be gained by factorising. We are now equipped to solve the PDE directly, and that is the approach taken.

With $\mathbf{X}_t$ and $\mathbf{Y}_t$ as before, birth joins the transition probabilities:

$$\begin{aligned} \Pr \{ \Delta \mathbf{X} = 1, \Delta \mathbf{Y} = -1 \mid \mathbf{Y}_t = j \} &= j \alpha \Delta t, & (\text{absorption}) \\ \Pr \{ \Delta \mathbf{X} = 1, \Delta \mathbf{Y} = 0 \mid \mathbf{X}_t = i \} &= i \beta \Delta t, & (\text{birth}) \\ \Pr \{ \Delta \mathbf{X} = -1, \Delta \mathbf{Y} = 0 \mid \mathbf{X}_t = i \} &= i \mu \Delta t, & (\text{death}) \\ \Pr \{ \Delta \mathbf{X} = 0, \Delta \mathbf{Y} = 0 \mid \mathbf{Y}_t = j, \mathbf{X}_t = i \} &= 1 - (j \alpha + i \mu + i \beta) \Delta t, & (\text{no event}) \end{aligned}$$

with forward equation

$$\frac{dp_{i,j}}{dt} = \alpha(j+1)p_{i-1,j+1} + \mu(i+1)p_{i+1,j} + \beta(i-1)p_{i-1,j} - (\alpha j + \mu i + \beta i)p_{i,j}. \quad (87)$$

The same reasoning as before gives the generating-function PDE

$$\frac{\partial G}{\partial t} = \left(\mu - (\mu + \beta)x + \beta x^2 \right) \frac{\partial G}{\partial x} + \alpha(x - y) \frac{\partial G}{\partial y}, \quad G(x, y, 0) = y^N. \quad (88)$$

The coefficient of $\partial G / \partial x$ is now quadratic rather than linear, which is the whole of the additional difficulty. Reparametrising $(x, y, t) \rightarrow (s, \rho, \tau)$ as in section 7.3 gives the characteristic system

$$\frac{dG}{d\tau} = 0, \quad (89)$$

$$\frac{dt}{d\tau} = -1, \quad (90)$$

$$\frac{dx}{d\tau} = \mu - (\mu + \beta)x + \beta x^2, \quad (91)$$

$$\frac{dy}{d\tau} = \alpha(x - y), \quad (92)$$

with the same initial conditions (78).

The equations for G and t , (89) and (90). Unchanged from before,

$$G(s, \rho, \tau) = \rho^N, \quad t = -\tau. \quad (93)$$

The equation for x , (91). This is a Riccati equation of the same shape as the one governing the survival probability in [section 4.4](#). Its right-hand side factorises, and it is worth noting that the roots are elementary rather than the surd expression the quadratic formula appears to promise:

$$\mu - (\mu + \beta)x + \beta x^2 = \beta(x - x_+)(x - x_-), \quad x_+ = 1, \quad x_- = \frac{\mu}{\beta}. \quad (94)$$

The discriminant is $(\frac{\mu+\beta}{2\beta})^2 - \frac{\mu}{\beta} = (\frac{\beta-\mu}{2\beta})^2$, a perfect square, so the surds cancel. It is convenient to name the combination that appears throughout,

$$b_1 = \beta(x_+ - x_-) = \beta - \mu, \quad (95)$$

the net interior growth rate. Separating variables in (91) and applying (78) gives the characteristic in the compact form

$$\frac{x - x_+}{x - x_-} = \left(\frac{s - x_+}{s - x_-} \right) e^{b_1 \tau}, \quad (96)$$

or, solved for x ,

$$x(\tau) = \frac{x_+(s - x_-) - x_-(s - x_+)e^{b_1 \tau}}{(s - x_-) - (s - x_+)e^{b_1 \tau}}. \quad (97)$$

The equation for y , (92). With an integrating factor the last characteristic equation becomes

$$\frac{d}{d\tau} (e^{\alpha \tau} y) = \alpha e^{\alpha \tau} x(\tau), \quad (98)$$

and the integral on the right is where the difficulty lies. Rather than substitute (97) and split the resulting quotient — which leads to two hypergeometric integrals and a good deal of bookkeeping — it is cleaner to expand $x(\tau)$ first. Writing $x - x_- = (x - x_+) + (x_+ - x_-)$ in (96) and solving for x gives

$$x(\tau) = x_+ + (x_+ - x_-) \frac{\zeta(\tau)}{1 - \zeta(\tau)}, \quad \zeta(\tau) = \left(\frac{s - x_+}{s - x_-} \right) e^{b_1 \tau}, \quad (99)$$

so that, for $|\zeta| < 1$,

$$x(\tau) = x_+ + (x_+ - x_-) \sum_{n=1}^{\infty} \zeta(\tau)^n.$$

Every term is now a pure exponential in τ and integrates immediately. Writing $Z = (s - x_+)/ (s - x_-)$, so that $\zeta(\tau) = Z e^{b_1 \tau}$ and $Z^n e^{nb_1 \tau} = \zeta^n$,

$$\begin{aligned} \alpha \int e^{\alpha \tau} x(\tau) d\tau &= \alpha x_+ \int e^{\alpha \tau} d\tau + \alpha(x_+ - x_-) \sum_{n=1}^{\infty} Z^n \int e^{(\alpha + nb_1)\tau} d\tau \\ &= e^{\alpha \tau} \left[x_+ + \alpha(x_+ - x_-) \sum_{n=1}^{\infty} \frac{\zeta^n}{\alpha + nb_1} \right] + \text{const.} \end{aligned}$$

Setting $\delta = \alpha/b_1$ and using $\alpha(x_+ - x_-)/b_1 = \alpha/\beta$ from (95), the bracket defines the function that carries the whole solution,

$$\Psi(\zeta) = x_+ + \frac{\alpha}{\beta} \sum_{n=1}^{\infty} \frac{\zeta^n}{n + \delta}, \quad \delta = \frac{\alpha}{b_1} = \frac{\alpha}{\beta - \mu}. \quad (100)$$

So $\alpha \int e^{\alpha\tau} x(\tau) d\tau = e^{\alpha\tau} \Psi(\zeta(\tau)) + \text{const}$, and integrating (98) gives $e^{\alpha\tau} y = e^{\alpha\tau} \Psi(\zeta(\tau)) + \text{const}$. The initial condition $y(s, \rho, 0) = \rho$ with $\zeta(0) = Z$ fixes the constant as $\rho - \Psi(Z)$, so

$$\rho = e^{\alpha\tau} [y - \Psi(\zeta(\tau))] + \Psi(Z). \quad (101)$$

Finally, put $\tau = -t$ and use (96): writing

$$W = \frac{x - x_+}{x - x_-}$$

we have $\zeta(\tau) = W$ and $Z = W e^{-b_1\tau} = W e^{b_1t}$. Substituting into (101) and applying (93) gives the solution.

Proposition 7.1 (Generating function for absorption–birth–death). *With $x_+ = 1$, $x_- = \mu/\beta$, $b_1 = \beta - \mu$, $\delta = \alpha/b_1$, $W = (x - x_+)/(x - x_-)$ and Ψ as in (100),*

$$G(x, y, t) = \left\{ e^{-\alpha t} [y - \Psi(W)] + \Psi(W e^{b_1 t}) \right\}^N. \quad (102)$$

Three checks confirm the result. At $t = 0$ the two Ψ terms cancel and $G = y^N$, the required initial condition. At $x = y = 1$ we have $W = 0$ and $\Psi(0) = x_+ = 1$, so $G = 1$ and probability is conserved. And setting $x = 1$ alone gives

$$G(1, y, t) = (e^{-\alpha t} y + 1 - e^{-\alpha t})^N,$$

so $\mathbf{Y}_t \sim \text{Binomial}(N, e^{-\alpha t})$ — exactly right, since the exterior particles are absorbed independently at rate α and what happens inside the compartment cannot affect them. Beyond these, (102) has been verified numerically against direct integration of the master equation (87), agreeing to around 10^{-13} across sub- and supercritical interior regimes and for $N = 1, 2, 3$.

Hypergeometric form

The series in (100) is a Gauss hypergeometric function in disguise. With

$${}_2F_1(a, b; c; z) = \sum_{n=0}^{\infty} \frac{(a)_n (b)_n}{(c)_n} \frac{z^n}{n!}, \quad (q)_n = \begin{cases} 1, & n = 0, \\ q(q+1) \cdots (q+n-1), & n \geq 1, \end{cases} \quad (103)$$

where $(q)_n$ is the rising Pochhammer symbol, the case $a = 1$, $b = \delta$, $c = \delta + 1$ collapses to a single ratio. Since $(1)_n = n!$ and

$$\frac{(\delta)_n}{(\delta+1)_n} = \frac{\delta(\delta+1) \cdots (\delta+n-1)}{(\delta+1) \cdots (\delta+n-1)(\delta+n)} = \frac{\delta}{\delta+n}, \quad (104)$$

we have

$${}_2F_1(1, \delta; \delta+1; z) = \sum_{n=0}^{\infty} \frac{\delta}{\delta+n} z^n. \quad (105)$$

Comparing with (100) and using $\alpha/(\beta\delta) = b_1/\beta = x_+ - x_-$,

$$\Psi(\zeta) = x_+ + (x_+ - x_-) [{}_2F_1(1, \delta; \delta+1; \zeta) - 1]. \quad (106)$$

(102) together with (106) is the closed form. It is not pretty, but it is a genuine closed form in terms of a standard special function, and the elementary series (100) is what one would actually evaluate. An alternative derivation of the same object, through the integral $\int e^{h\tau} / (P + Qe^{k\tau}) d\tau$, is given in [appendix A](#); that route is longer, but it produces a general identity of independent use.

Remark 7.2 (Domain of validity). The series (100) converges for $|\zeta| < 1$, and the arguments appearing in (102) are W and $We^{b_1 t}$. In the subcritical interior regime $\mu > \beta$ we have $b_1 < 0$ and $x_- = \mu/\beta > 1$, so for $x \in [0, 1]$ the argument W lies in $[0, \beta/\mu] \subset [0, 1)$ and $We^{b_1 t}$ is smaller still: the series representation is valid on the whole of the unit square, which is the case of most interest here. When $\beta > \mu$ the series converges only for x above the smaller fixed point μ/β , and elsewhere one must use the analytic continuation of ${}_2F_1$ — conveniently, the Euler integral

$${}_2F_1(1, \delta; \delta + 1; \zeta) = \delta \int_0^1 \frac{u^{\delta-1}}{1 - \zeta u} du \quad (\delta > 0, \zeta \notin [1, \infty)),$$

which was used for the numerical verification quoted above.

Remark 7.3 (Resonance). The denominators in (100) vanish when $n + \delta = 0$ for some integer $n \geq 1$, that is when

$$\alpha = n(\mu - \beta), \quad n = 1, 2, \dots$$

This is the classical resonance condition for linearising the y equation: the homogeneous decay rate α of the absorption channel coincides with a harmonic of the interior relaxation rate $\mu - \beta$. The generating function itself is perfectly regular there, being analytic in all parameters, so the singularity is removable, and at resonance the corresponding term acquires a factor τ — a secular term — in place of the simple exponential. The resonant parameter values form a set of measure zero and the limit may be taken numerically, but the degeneracy is worth keeping in mind when α happens to sit near an integer multiple of $\mu - \beta$.

8 What carries forward

Three things assembled here are used repeatedly in the chapters that follow.

The first is a working method rather than a result. A model is specified by rates; the rates give a generator and a master equation; multiplying by z^i (or by $x^i y^j$) and summing converts that master equation into a single first-order partial differential equation for a generating function; and the method of characteristics of [section 7.3](#) solves equations of that type. (102) shows the route completed on a problem for which no shortcut exists, and the same route is followed in later chapters when a catastrophe channel is added to the rates of (14). [Appendix B](#) supplies the last step, recovering time-dependent state probabilities from a generating function given by a formula rather than a series.

The second is the behaviour of small populations near criticality. Conditioning on survival converts a process whose mean is uninformative into one with a stable conditional distribution, and for the binary Galton–Watson process the whole of that distribution’s first two moments is controlled by $A(p)$. The constant admits an exact product (51), an exact series (52), two-sided bounds (53) that place it between $\frac{1}{2}A_c(p)$ and $2\varepsilon/(1 + 2\varepsilon)$, and the near-critical asymptotic $A(p) \sim 2(1 - 2p)$. It does not appear to admit an elementary closed form, and [section 6.6](#) gives the structural reason to expect none. What it does admit is a cheap and accurate hybrid scheme, (68), which is what the numerics of later chapters use.

The third is a set of specific questions left open here and taken up later. Whether a quasi-stationary distribution survives when a birth–death–catastrophe process is conditioned on *neither* extinction nor rupture is not settled by the argument of [section 5.4](#), and is best approached by simulation. How tight the Jensen bound (44) is remains a numerical question, and a sharp answer would make the discrete catastrophe calculation usable in its own right. And whether the map $p \mapsto A(p)$, as opposed to the Koenigs function ψ_r in its argument, is elementary is a transcendence question about values that [section 6.6](#) does not answer.

Appendices

A An integral identity for the hypergeometric function

The route to (100) taken in section 7.4 expands the characteristic $x(\tau)$ in a geometric series and integrates term by term. An alternative is to integrate the quotient (97) directly, splitting the numerator into two pieces, each of the form

$$I(\tau) = \int \frac{e^{h\tau}}{P + Qe^{k\tau}} d\tau, \quad (107)$$

with h, k, P, Q constant. That route is longer and easier to get wrong — the two pieces carry different prefactors, and dropping one produces a formula that still satisfies the initial condition while violating $G(1, 1, t) = 1$ — but the identity it rests on is useful in its own right, so we record and prove it here.

Proposition .1. *For $h/k > 0$ and $-\frac{Q}{P}e^{k\tau} \notin [1, \infty)$,*

$$\int \frac{e^{h\tau}}{P + Qe^{k\tau}} d\tau = \frac{e^{h\tau}}{Ph} {}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{Q}{P}e^{k\tau}\right) + \text{const.} \quad (108)$$

Proof. We use the Euler integral representation of the hypergeometric function,

$${}_2F_1(a, b; c; z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 u^{b-1}(1-u)^{c-b-1}(1-zu)^{-a} du, \quad (109)$$

Γ being the standard gamma function.

Start from (107) and substitute $v = e^{k\tau}$, so that $d\tau = k^{-1}v^{-1}dv$:

$$I = \frac{1}{k} \int \frac{v^{h/k}}{P + Qv} v^{-1} dv = \frac{1}{Pk} \int \frac{v^{h/k-1}}{1 + \frac{Q}{P}v} dv.$$

Fix the antiderivative by writing this as a definite integral from 0 to v in a dummy variable w ; the integral converges at the lower limit precisely because $h/k > 0$:

$$I = \frac{1}{Pk} \int_0^v \frac{w^{h/k-1}}{1 + \frac{Q}{P}w} dw.$$

Now rescale, $w = vu$, so that $dw = v du$ with v held fixed:

$$I = \frac{1}{Pk} \int_0^1 \frac{(vu)^{h/k-1}}{1 + \frac{Q}{P}vu} v du = \frac{v^{h/k}}{Pk} \int_0^1 \frac{u^{h/k-1}}{1 + \frac{Q}{P}vu} du. \quad (110)$$

The remaining integrand matches that of (109),

$$\int_0^1 u^{h/k-1}(1-u)^0 \left(1 - \left(-\frac{Q}{P}v\right)u\right)^{-1} du = \int_0^1 u^{b-1}(1-u)^{c-b-1}(1-zu)^{-a} du,$$

with $b = h/k$, $c = h/k + 1$, $a = 1$ and $z = -\frac{Q}{P}v$. Hence, using $\Gamma(z+1) = z\Gamma(z)$,

$$\int_0^1 \frac{u^{h/k-1}}{1 + \frac{Q}{P}vu} du = \frac{\Gamma\left(\frac{h}{k}\right)\Gamma(1)}{\Gamma\left(\frac{h}{k} + 1\right)} {}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{Q}{P}v\right) = \frac{k}{h} {}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{Q}{P}v\right).$$

Substituting into (110),

$$I = \frac{v^{h/k}}{Ph} {}_2F_1\left(1, \frac{h}{k}; \frac{h}{k} + 1; -\frac{Q}{P}v\right),$$

and restoring $v = e^{k\tau}$ gives (108). $\square$

Remark .2. Applied to the two pieces of (97) — the first with $h = \alpha$, the second with $h = \alpha + b_1$, and both with $k = b_1$, $P = (s - x_-)$, $Q = -(s - x_+)$ — proposition .1 reproduces (106) once the prefactors $\alpha x_+(s - x_-)$ and $-\alpha x_-(s - x_+)$ have been carried through correctly. The factor $(s - x_+)/(s - x_-)$ attached to the second piece is easily lost, and its loss is invisible at $t = 0$; checking $G(1, 1, t) = 1$ catches it immediately.

B Extracting state probabilities from a generating function

A generating function defined by a formula rather than as an explicit series — for example (73) — can always be converted back to the form

$$G(x, y, t) = \sum_{i=0}^{\infty} \sum_{j=0}^{\infty} p_{i,j}(t) x^i y^j, \quad (111)$$

recovering the time-dependent state probabilities. All that is required is that G be repeatedly partially differentiable in x and y .

The reason is the bivariate Maclaurin expansion,

$$f(x, y) = \sum_{k=0}^{\infty} \frac{1}{k!} \sum_{\varrho=0}^k \binom{k}{\varrho} \partial_x^{\varrho} \partial_y^{k-\varrho} f \Big|_{(0,0)} x^{\varrho} y^{k-\varrho},$$

whose coefficient of $x^i y^j$, obtained by taking $k = i + j$ and $\varrho = i$, is $\frac{1}{(i+j)!} \binom{i+j}{i} \partial_x^i \partial_y^j f \Big|_{(0,0)}$. Hence

$$p_{i,j}(t) = \frac{1}{i! j!} \partial_x^i \partial_y^j G \Big|_{(0,0)}. \quad (112)$$

The multilinear case

Both (73) and (77) have the form

$$G(x, y, t) = (c(t) + f(t)x + g(t)y)^n, \quad (113)$$

for which the derivatives are immediate. Each differentiation lowers the power by one and brings down a factor, so

$$\partial_x^a \partial_y^b G = \frac{n!}{(n-a-b)!} f^a g^b (c + fx + gy)^{n-a-b} \quad (a + b \leq n), \quad (114)$$

and zero for $a + b > n$. Evaluating at the origin and applying (112),

$$p_{i,j}(t) = \frac{n!}{i! j! (n-i-j)!} f(t)^i g(t)^j c(t)^{n-i-j}, \quad i + j \leq n, \quad (115)$$

and $p_{i,j} = 0$ otherwise. This is the trinomial distribution, which is exactly what it should be: the n particles are independent, and at time t each is in one of three states with probabilities c (neither inside nor outside — that is, dead), f (inside) and g (outside). Note in particular that the falling factorial $n!/(n-i-j)!$ in (114) is *not* the same as n^{i+j} ; the two agree only to leading order in n , and using the latter breaks normalisation.

Applied to the absorption models

For absorption only, (73), we have $c = 0$, $f = 1 - e^{-\alpha t}$ and $g = e^{-\alpha t}$. The factor c^{n-i-j} then vanishes unless $i + j = n$, and (115) gives

$$p_{i,j}(t) = \binom{n}{i} (1 - e^{-\alpha t})^i (e^{-\alpha t})^j, \quad i + j = n,$$

so $\mathbf{X}_t \sim \text{Binomial}(n, 1 - e^{-\alpha t})$ and $\mathbf{Y}_t = n - \mathbf{X}_t$, as anticipated in section 7.1.

For absorption–death, (77), the three functions are

$$c(t) = 1 - \frac{\mu}{\mu - \alpha} e^{-\alpha t} + \frac{\alpha}{\mu - \alpha} e^{-\mu t}, \quad f(t) = \frac{\alpha}{\mu - \alpha} (e^{-\alpha t} - e^{-\mu t}), \quad g(t) = e^{-\alpha t},$$

which are precisely $p_{0,0}$, $p_{1,0}$ and $p_{0,1}$ of (76), and

$$p_{i,j}(t) = \frac{n!}{i! j! (n - i - j)!} f(t)^i g(t)^j c(t)^{n-i-j}.$$

Each particle independently ends up dead, absorbed, or still outside, with those three probabilities, and the joint law is trinomial. That the generating function factorises in this way is (72) read backwards.

The absorption–birth–death generating function of proposition 7.1 is not of the form (113), and there (112) must be applied to (102) directly, or the coefficients extracted numerically by Cauchy integrals on circles $|x| = |y| = \varrho < 1$.

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